LandSlide Segmentation Using Deep Learning Models

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CERTIFICATE

This is to certify that this project report titled **"LANDSLIDE SEGMENTATION USING DEEP LEARNING MODELS"** is a bonafide record of work done by **VATSALYA GAYATRI CHALAVADI (228W1A1211), PALLAVI NIMMANAGOTI (228W1A1242)** under my guidance and supervision is submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Information Technology, **V.R. Siddhartha Engineering College** (Autonomous under JNTUK) during the year 2025-26.

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PROJECT SUMMARY

S.No	Item	Description
1	**Project Title**	**LANDSLIDE SEGMENTATION USING DEEP LEARNING MODELS**
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4	**Research Group**	**Disaster Management and Environmental Monitoring**
5	**Application Area**	**Landslide early warning systems**
6	**Aim of the Project**	**To develop a transformer based framework for accurate and automated landslide segmentation using satellite imagery, leveraging advanced architectures**
7	**Project Outcomes**	**A fully trained deep learning model capable of distinguishing landslide areas from non-landslide areas in satellite images.**

Student Signatures

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Signature of the Guide

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ABSTRACT

Landslides are the most devastating natural hazards, which are life threatening, destructive to infrastructure and to the environment. The conventional techniques of landslide detection through manual reading of remote sensing data are time-consuming, subjective and ineffective in covering a large area. This paper aims to tackle these issues by suggesting an automated deep learning-based model of precise landslide segmentation through satellite images in the Landslide4Sense dataset. The images undergo a preprocessing step and are scaled to 256x256 pixels to make them consistent and model rapid training. The system suggested tries and evaluates various segmentation models, such as U-Net, DDR U-Net, Attention U-Net, SegFormer, and MobileViT. CNN-based models are good at capturing the spatial characteristics and avoiding boundaries, but transformer-based and hybrid models are more effective to understand the global context and achieve better segmentation performance in tricky terrains. The models are compared in terms of pixel-wise measures: Accuracy, Precision, Recall, F1-score, IoU and Dice coefficient. Moreover, the framework incorporates area estimation and severity classification to determine the impact of landslides. The experimental findings indicate that sophisticated models, especially transformer-based models, offer superior feature representation and segmentation performance and the system is applicable in real-time monitoring, disaster management and geospatial risk evaluation.

CHAPTER-1
INTRODUCTION

One of the most devastating and the most common natural calamities is landslides that leave a devastating impact on human life, infrastructure and the environment. Heavy rains, soil erosion, slope instability as well as earthquakes normally cause them. Early detection and monitoring of landslides are significant to minimize the losses and make disaster management timely. Landslides have conventionally been determined by the manual analysis of remote sensing pictures. Although such an approach can provide accurate results, it is time-consuming, cumbersome and may be subject to human mistakes [1].

As landslides continue to occur, especially in hilly and densely populated areas, the necessity to have automated, efficient and reliable landslide detection systems is increasing. The latest evolution of remote sensing technologies, as well as the fast progress of Artificial Intelligence (AI) and deep learning, have greatly enhanced the capabilities of analyzing natural hazards. Widely used in image segmentation tasks, [2] Convolutional Neural Networks (CNNs) have been shown to effectively extract meaningful features in high-resolution images. The U-Net architecture is one of the most popular among them, as it has an encoder-decoder architecture, skip connections that allow maintaining spatial and contextual information. The traditional U-Net models might not however be successful in capturing fine boundaries as well as complex features in heterogeneous terrains, particularly when there is different lighting conditions and vegetation cover. This has been curtailed by the development of more feature extracting and more performing sophisticated architectures with regard to segmentation.

1.1 ORIGIN OF THE PROBLEM

Landslides are the most common and dangerous natural catastrophes especially in those areas where urbanization, deforestation, unplanned construction and extreme weather conditions like heavy rains and earthquakes have occurred. Such incidences cause serious impacts such as loss of human lives, damaged infrastructure like roads, bridges, and structures, transport system disruption, economic losses and environmental degradation in the long term. The growing effects of climate change have also increased the frequency and unpredictability of landslides making it a significant issue to disaster management authorities across the globe.[2]

In the past, landslide detection and monitoring has been through manual methods like field surveys, geological mapping and satellite imagery visual interpretation. Despite the potential to give precise information, these approaches are time-consuming, labor-intensive, and need

expert knowledge. Additionally, they cannot be scaled to large geographic areas and are usually not practical in remote or inaccessible areas. Subsequently, late responses may cause more damage in terms of delayed response and a decrease in disaster mitigation measures due to delay in detection.

High-resolution satellite images have become easily accessible with the swift development of remote sensing technologies, which can give a detailed overview of the terrain, vegetation, and environmental changes. Nevertheless, it is still a major challenge to extract meaningful insights out of such large volumes of data using conventional methods. [1] This has given rise to using Artificial Intelligence (AI) and deep learning-based methods that can automatically examine intricate patterns in space and detect areas of landslides more efficiently and accurately. Previous deep learning architectures like U-Net and its variations have demonstrated encouraging performance in landslide segmentation through the use of convolutional neural networks (CNNs). Nevertheless, these models are mainly concerned with local feature extraction and may not be able to extract global contextual information, which is essential in the interpretation of large-scale terrain variations. This is a weakness that impacts their capability to observe the presence of landslides in complicated and non-homogenous terrain.

The new development in transformer-based architectures, including SegFormer and hybrid architectures like MobileViT, is a more potent solution to address these challenges. These models can obtain both local and global relationships in an image allowing better representation of features and more precise segmentation. With the combination of these sophisticated models and the remote sensing data, one can come up with an automated, efficient and scalable system of landslide detection.

1.2 BASIC DEFINITIONS AND BACKGROUND

1.2.1 Landslide:

A form of natural disaster, which entails downward movement of rock, earth or debris along a slope. It may occur naturally through rainfalls, earthquakes or manmade activities like construction and mining[1].

1.2.2 Segmentation:

Segmentation is a computer vision method which labels each pixel in an image with a class label. When analyzing landslides each pixel is labelled as landslide or non-landslide. This pixel-wise prediction helps in delineating the exact shape, size, and boundary of landslide regions and is essential for accurate geospatial interpretation[1].

1.2.3 U-Net Architecture:

U-Net is an encoder-decoder neural network that is commonly applied in medical and remote sensing segmentation. Convolution and pooling are used to extract image features in the encoder. The decoder recreates the segmentation mask. Skip connections carry fine details between encoder and decoder enhancing boundary preservation [1].

1.2.4 Deep Deconvolutional Residual U-Net (DDR U-Net):

DDR U-Net is a better U-Net model, which uses Residual blocks, which, in turn, ensures preserving the gradient flow and avoiding vanishing gradients and Deconvolutional layers, which further details the recovery of spatial details during upsampling [1].

1.2.5 Attention U-Net:

Attention U-Net is an extension of U-Net, which adds Attention Gates (AGs). AGs automatically prioritize on the areas of highest interest (landslide areas) and reduce background noise. This enhances accuracy in segmentation especially on small or obscure landslides. [1].

1.2.6 Activation Functions:

The three models involved in this project U-Net, DDR U-Net, and Attention U-Net are mainly based on the ReLU (Rectified Linear Unit) activation function in their convolutional layers to effectively extract spatial information, and prevent vanishing gradients. The last output layer of every model will be the binary segmentation mask generated by the sigmoid activation function which will mark each pixel as landslide or non-landslide.[3].

1.2.7 SegFormer:

SegFormer is a hierarchical encoder-based transformer-based semantic segmentation model that uses a lightweight decoder. The encoder derives multi-scale features with the help of self-attention and the decoder effectively combines these features to produce correct segmentation masks. It is created in such a way that it can deal with different sizes of images and complicated terrains without much dependence on positional encoding [2].

1.2.8. MobileViT:

MobileViT is a hybrid deep learning model that integrates convolutional layers with lightweight transformer blocks. It captures local spatial features using CNNs and global contextual information using transformers. This architecture provides a balance between accuracy and computational efficiency, making it suitable for real-time and resource-constrained applications [2].

1.2.9 Patch Embedding:

Patch embedding is a transformer model process where a picture is broken down into smaller patches and transformed to a sequence of feature vectors. This enables the model to process images like sequences which helps it to make good use of attention mechanisms [2].

1.2.10 Intersection over Union (IoU):

IoU is a performance measure of segmentation problems to quantify how much the predicted and ground truth regions overlap. It is given by a ratio between the area of intersection to the sum of the predicted and actual regions [3].

1.2.11 Dice Coefficient:

The Dice coefficient is a similarity measure that is used to measure segmentation. It compares the predicted regions to the actual ones and assigns greater weight to the pixels that are correctly predicted [3].

Background:

Landslides are among the greatest geohazards with a considerable loss of human life, infrastructure damage, and a lasting effect on the environment. The rising rate of climate change, deforestation, and rapid urbanization that have led to the rising number of landslides have rendered early detection and monitoring of landslides as of paramount importance in disaster preparedness and mitigation of risks. Conventional landslide mapping techniques are greatly dependent on manual identification of remote sensing data and field mapping. Though they may be accurate, these methods are time consuming, subjective and cannot be used in large scale or real time monitoring, particularly in inaccessible and remote areas. As remote sensing technologies have improved, high-resolution satellite images are now widely accessible, providing an in-depth spatial and temporal data of terrain conditions. Nevertheless, it is still difficult to derive meaningful insights on such huge amounts of data with traditional methods. Deep learning has become one of the potent solutions to automated image analysis in the recent years, especially in semantic segmentation problems. Convolutional Neural Networks (CNNs) with architectures like U-Net, DDR U-Net and Attention U-Net have demonstrated high effectiveness in the extraction of spatial features, boundary preservation, and enhancement of concentration on pertinent parts of a heterogeneous landscape. Although effective, CNN-based models are mostly local in nature, and they might not be able to extract global contextual information, which is critical to interpreting large-scale terrain variability and complicated landslide patterns [1]. To overcome these shortcomings, recent studies have

proposed transformer based and hybrid architectures like SegFormer and MobileViT. SegFormer employs an encoder made up of transformers to extract long-range dependencies and the global context so that it can understand larger and complex region[2]. MobileViT is based on convolutional operations and lightweight transformer blocks, which enables it to effectively learn local and global features and remain computationally efficient. With these developments, this paper will create an automated and robust landslide segmentation model that utilizes CNN-based and transformer-based methods. Combining SegFormer and MobileViT with preprocessing methods, pixel-wise segmentation, area estimation, and severity classification, the suggested system offers a complete system, which can accurately detect landslides. This method improves the power of geospatial analysis and the real-life uses including early warning systems, disaster management and risk assessment.

1.3 PROBLEM STATEMENT

Landslides are highly dangerous to the environment, infrastructure and human life, although the accurate and timely detection of landslides remains a major challenge. Conventional techniques of remote sensing image interpretation that rely on manual interpretation are slow, subjective and not applicable to large scale or real-time monitoring in disaster prone areas. Though the recent development of deep learning models like the U-Net and its variants have enhanced automated landslide segmentation, they are limited in capturing complex boundaries, small or irregular landslide localities, and maintaining global contextual information in heterogeneous landscapes. Specifically, the traditional convolutional neural network (CNN) models mainly concentrate on local features extraction and might be unable to grasp long-range spatial interactions, which are vital to the successful segmentation of big and complicated landslide regions. This weakness impacts their operation in various geographical settings where patterns of terrain differ greatly [1].

To address these issues a more solid and efficient segmentation model is required that is capable of capturing local and global attributes effectively. The proposed study will enhance both the accuracy and reliability of landslide identification of remote sensing images through the fusion of cutting-edge transformer-based and hybrid frameworks, namely SegFormer and MobileViT. These models improve feature representation by use of self-attention processes and effective feature fusion to allow more accurate pixel-level segmentation [2].

In addition, the research goes beyond segmentation to include area estimation and severity classification, which are valuable insights to the magnitude and consequences of landslides.

The overall idea of this approach is to create an automated landslide detection system that is scalable and reliable enough to provide real-time monitoring, early warning systems and geospatial risk assessment under various environmental pressures [4].

1.3.1 Objectives:

- Create a deep learning-based landslide segmentation framework based on satellite images through the integration of advanced preprocessing methods to boost the visibility and consistency of features.
- Apply and compare transformer-based and hybrid architectures, namely SegFormer and MobileViT, to detect landslides precisely and pixel-wisely.
- Apply and evaluate transformer-based and hybrid designs, namely SegFormer and MobileViT, to detect landslides on a pixel-by-pixel basis correctly.
- Train and compare the transformer-based and hybrid architectures, namely SegFormer and MobileViT, to detect landslides in pixels accurately.

1.3.2 Outcomes:

- Trained and deployed SegFormer and MobileViT models on the Landslide4Sense dataset with images of resized size 256×256 with success.
- Realized better segmentation performance because of better grounding of local spatial features and world context information especially in complex terrain conditions.
- Noted that MobileViT was very accurate and balanced, and SegFormer was very effective in global dependencies of satellite imagery.
- Better overall segmentation performance over simple models, and enhanced boundary detection, feature representation and minimized misclassification of background pixels.
- Effectively combined area estimation and severity classification, which allowed the system to present valuable information to be used in landslide risk assessment and disaster management uses.

1.4 SOCIETAL APPLICATIONS

The automated landslide detection system by deep learning suggested in this paper has an extensive list of real-world applications:

- **Disaster Management and Response:** Allows quick detection of areas of landslides to evacuate and provide relief promptly. Helps to design safe access routes in case of emergencies.

- **Early Warning Systems:** It is integrated with real-time monitoring systems and early alerts are provided before landslides happen and casualties and damage of property can be minimized.
- **Transportation Safety:** Helps to monitor highways, railways, and hill roads in case of a possible landslide and enhances the safety of traveling and decreases accidents.
- **Insurance and Risk Assessment:** Assists insurance companies in determining the risk levels and estimating the possible damage, which results in improved policy planning and claim management.
- **Climate Change Impact Analysis:** Assists with the examination of the effects of climate change on landslide frequency, based on the examination of rainfall patterns, changes in the terrain, and vegetation loss.
- **Urban and Infrastructure planning:** Helps planners do not build buildings, roads or any other infrastructure in areas that are prone to landslides. Favors long-term urban development plans in high hazard areas.
- **Environmental Monitoring:** Measures the terrain, vegetation and rainfall variations to forecast the occurrence of landslides. Monitors the impact of human activities such as deforestation and mining.
- **Remote and Rural Area Monitoring**: Allows the monitoring of inaccessible or remote mountainous locations where a manual inspection is challenging or impossible.
- **Geospatial Data Integration:** Improves Geographic Information Systems (GIS) that are employed by government and researchers. Allows automated analysis of geotechnical data.

CHAPTER –2

REVIEW OF LITERATURE

This chapter describes the review of literature that we have taken from various papers and considered all the points mentioned in the papers.

2.1 DESCRIPTION OF EXISTING SYSTEMS:

Table 2.1: Literature Review

S.NO	Paper title	Year	Datasets, algorithms	Results	Limitations
1.	LandslideSegNet: an effective deep learning network for landslide segmentation using remote sensing imagery.*(Earth Science Informatics)*	2024	Datasets: RapidEye,Sentinel-1 ,Sentinel-2, Landsat8,GVLM Algorithms: LandSlideSegnet,Unet,PSPNet, DeepLabV3,Enet,FCN8,ICNet.	LandslideSegNet achieved 97.6% accuracy and outperforming U-Net, PSPNet, DeepLabV3+ and SegNet	Struggles with precise delineation of landslide edges, especially in complex terrain
2.	Potential of Synthetic Images in Landslide Segmentation in Data-Poor Scenario: A Framework Combining GAN and Transformer Models	2024	Datasets: Generated using StyleGAN Algorithms: CNN models (UNet,DeepLabV3+, FCN, PSPNet, ConvNeXt), Transformer Models (Swin SegFormer, VIT)	Adding StyleGAN2 synthetic images improved segmentation accuracy by over 10% for most models transformer models showed highest performance.	Synthetic images still require manual labelling, and GAN-based data lacks automatic mask generation, making the process diffcult
3.	Semantic segmentation of loess landslides with STAPLE mask and fully connected	2023	Dataset: Loess Landslide Dataset Algorithm: U-Net + STAPLE + Fully Connected CRF	Achieved highest Dice score and improved boundary accuracy compared to baseline models	Requires multi-expert labels, dataset is small & region-specific, cannot generalize or predict future

					landslides
4.	Detection and segmentation of loess landslides via satellite images: a two-phase framework	2022	Dataset: loess landslide satellite images Algorithms: R-CNN,Vanilla U-Net, U-Net+CRF, U-Net + data fusion	R-CNN, Achieved a Dice Similarity Coefficient (DSC) of 0.68 and outperformed vanilla U-Net, FCN, and U-Net+CRF in segmentation accuracy.	Low contrast and weak boundaries of loess terrain reduce segmentation accuracy.
5.	Efficient and automatic extraction of slope units based on multi-scale segmentation method for landslide assessments	2021	Dataset: DEM data of Chongyi County. Algorithms: Multi-Scale Segmentation (MSS) + GIS Hydrological Method	MSS method extracted slope units more accurately, than the hydrological method and better matched real topography and landslides.	The approach lacks integration of geological and hydrological variables, reducing its robustness.
6.	Landslide Inventory Mapping From Bitemporal High-Resolution Remote Sensing Images Using Change Detection and Multiscale	2018	Change Magnitude Image (CMI) Multithresholds to generate binary maps Majority Voting (MV) to combine binary maps of post-event image Object-Based Voting (OBV) Pre- and post-event high-resolution aerial images	Area A:90.7 AreaB:80.1 Area C:60.06	Lower Accuracy in Forested Areas: Sensitivity Noise Sensitivity in High-Resolution Images

7.	Attention U-Net for Landslide Detection Using Remote Sensing Data	2021	Sentinel-2 satellite images, Landslide inventory datasets Attention U-Net, CNN with attention gates	Attention mechanism improved feature selection and enhanced segmentation accuracy compared to standard U-Net. The model showed better focus on relevant landslide regions and reduced false positives.	Still struggles with small-scale landslides and requires careful tuning of attention parameters.
8.	SegFormer: Simple and Efficient Transformer for Semantic Segmentation Applied to Landslides	2022	Sentinel-1, Sentinel-2, Landslide4Sense dataset, SegFormer (Transformer-based), Mix Vision Transformer (MiT), MLP Decoder	SegFormer demonstrated superior performance over CNN-based models by capturing global context efficiently. It achieved higher mIoU and better generalization across different terrains.	Transformer models require high memory and training data; performance depends heavily on dataset quality.
9.	DeepLabv3+ for Semantic Segmentation of Landslides Using High-Resolution Remote Sensing Images	2020	High-resolution satellite imagery (Google Earth, UAV images) DeepLabv3+, Atrous Spatial Pyramid Pooling (ASPP), Encoder-Decoder CNN	DeepLabv3+ achieved high segmentation accuracy with improved boundary detection compared to FCN and U-Net. The use of ASPP helped capture multi-scale contextual information effectively.	Requires large labeled datasets and high computational resources; performance drops in highly heterogeneous terrain.

2.2 SUMMARY OF LITRATURE REVIEW

Abdullah The work of Abdullah Şener and Burhan Ergen is an effective deep learning network that can be used to detect landslides, namely LandslideSegNet: an effective deep learning network to detect landslides with remote sensing images [1], which suggests a special deep learning architecture that is dedicated to the task of image-based landslide detection. The model is developed in encoder-decoder format with the incorporation of Encoder-Decoder Residual (EDR) blocks to enhance feature propagation and to avert the loss of information in training. Also, the Efficient Hybrid Attentional Atrous Convolution (EHAAC) is integrated to allow the network to extract local and global contextual information, with multi-scale receptive fields. Such combination enables the model to be successful in detecting the areas of landslides even in complicated terrain conditions. The model was trained and tested on a dataset (Landslide4Sense) that consists of various satellite images using different sensors. It was shown that LandslideSegNet has a high accuracy of 97.6% as well as a high mean Intersection over Union (mIoU), which is better than popular networks like U-Net, PSPNet, DeepLabV3+, and FCN. The lightweight nature of the model also guarantees lower computational complexity which means that it can be used in real-time or near real-time applications. Nonetheless, the model has been challenged in its ability to draw precise boundaries of landslides especially where the land cover and topography is irregular. Moreover, the lack of transfer learning also restricts its extrapolation to other datasets, which presents a crucial aspect of future improvement.

Xiao Feng et al. introduced the article, Potential of Synthetic Images in Landslide Segmentation in Data-Poor Scenario [2], that addresses the problem of a lack of labeled data to identify landslides. The authors suggested a hybrid model combining StyleGAN2 to generate synthetic images with both a convolutional neural network (CNNs) and a transformer-based segmentation model. The synthetic images increase the size of the dataset and enable models to learn a wider range of more representative features. Several architectures (U-Net, DeepLabV3+, FCN, PSPNet, ConvNeXt, and also transformer-based models like Swin Transformer, SegFormer, and Vision Transformer (ViT) were evaluated in the study. The findings revealed that the accuracy of segmentation increased by over 10-15 percent using the synthetic data in most of the models. Transformer-based models performed better because they can acquire long-range dependencies and contextual relationships in images on a global scale. This indicates how data augmentation methods together with advanced architectures are vital in enhancing the robustness of models.

The study also, however, lists a few limitations. Synthetic images produced by GANs need to have segmentation masks manually annotated, which makes them more costly to prepare the dataset. Also, the quality of synthetic data might be not a complete copy of the real-world conditions, which might cause biases in the model. Nevertheless, the framework offers a good prospect to such data-scarce situations and emphasizes the increased significance of generative models within the context of remote sensing.

Huajin Li et al. suggested Semantic segmentation of loess landslides with STAPLE mask and fully connected conditional random field [3], where the aim is to enhance the reliability of the segmentation in areas with high uncertainty in the annotation. Landslide datasets have the problem of poor consistency in their labeling because of subjective interpretation of the data by various professionals. To solve this problem, the authors proposed the STAPLE (Simultaneous Truth and Performance Level Estimation) algorithm that takes into consideration several expert annotations in order to produce a probabilistic and much more accurate ground truth mask. This will greatly decrease bias and enhance the quality of training data. The foundational segmentation model in this paper is U-Net that is famous with its encoder-decoder architecture and performance in pixel-wise classification problems. The authors also added a Fully Connected Conditional Random Field (FC-CRF) post-processing step to improve the outcomes of the segmentation further. The CRF takes into account the spatial relations between pixels and refines the predicted segmentation map, enhancing the boundary accuracy and minimizing noise. This use of STAPLE and CRF assists in getting more precise and refined outputs in segmentation. The results of the experiment proved that the given method has reached a Dice coefficient of about 0.80, which is higher than the classical U-Net-based methods. The framework also facilitated quantitative research of landslide morphology, which can be of use in geological research. The cost and complexity of dataset preparation however increases because of the need to have several expert annotations. Also, the data utilized is quite small and confined to particular loess areas, limiting the extrapolation power of the model. Future recommendations involve using larger datasets and transfer learning to make it more robust.

Huajin Li et al. presented the article Detection and segmentation of loess landslides via satellite images: a two-phase framework [4], whose purpose is to enhance the landslides detection by splitting the detection and segmentation into two phases. The model applies Faster R-CNN, a region based object detection algorithm in the first stage to detect the possible regions of landslides in satellite images. The step aids in the process of refining the areas of interest and minimizes the needless calculations during the segmentation. The second phase involves

sending the detected regions to an improved U-Net model with inception modules. These modules enable the network to detect features at different scales which is required in detecting landslides of different sizes and shapes. Detection and segmentation are more effectively integrated to enhance the localization and refining the boundary, resulting in improved performance. The framework had a Dice Similarity Coefficient (DSC) of 0.68, which is better than baseline models like FCN and regular U-Net. The two stage process is also more efficient since it allocates computational power to the only pertinent areas. The model however has difficulties in low-contrast conditions especially in loess terrains where landslides and the surrounding terrain have similar appearances.

Also, overfitting is caused by the small size of the training dataset, which decreases the capacity of the model to be used in new areas. As noted in the study, more varied datasets and better methods of feature extraction are required to deal with complex terrains. The paper by Faming Huang et al. offers an alternative approach to slope unit extraction and multi-scale segmentation method to estimate landslides with a specific emphasis on the terrain-based segmentation instead of image-based segmentation [5]. The research makes use of Digital Elevation Model (DEM) data to isolate slope units, which are critical in the study of the landslide propensity and the topography.

The Multi-Scale Segmentation (MSS) approach suggested breaks down the terrain into meaningful units, based on the elevation, slope, and hydrological features. This methodology is such that the units of slope extracted are similar to actual topography. It was experimented using data of Chongyi County and Wanzhou District, and it showed a better morphological consistency and closer alignment with the real landslide patterns than the conventional hydrological approaches. Computational efficiency is one of the benefits of this approach and it is applicable in large-scale applications. Nonetheless, the technique mainly depends on geographical characteristics and fails to consider other significant variables like soil characteristics, vegetation cover and climatic variations. Also, it does not have deep learning techniques, which restrict its capacity to learn complicated forms of data. The combination of MSS and machine learning/deep learning models may also be used to improve its efficiency and implementation in practical situations. The article Landslide Inventory Mapping on Bitemporal High-Resolution Remote Sensing Image Change Detection and Multiscale Segmentation [6] introduces a technique to locate landslides based on the difference between pre-event and post-event photos. The method relies on the belief that landslides result in considerable alterations in surface features, which can be identified using image differencing methods. It employs Change

Magnitude Imaging (CMI) to identify regions of great change, and then multi-threshold segmentation to produce binary maps. Very Simple Majority Voting (MV) and Object-Based Voting (OBV) are then used to combine these maps to enhance the accuracy and minimize noise. Multi-methodology provides a powerful detection of the landslide areas.

Experimental Findings demonstrated that the technique was quite accurate in various test regions, where the value was 90.7, 80.1 and 60.06. Although these are encouraging findings, there are a few limitations of the technique. It has a low performance in forested areas where landslide features are not visible due to vegetation cover and thus not easy to detect. Also, the method is sensitive to noise in high-resolution pictures and needs both pre-event and post-event data, which are not always available.

This limits its applicability for real-time monitoring and prediction. DeepLabv3+ is a more sophisticated convolutional neural network framework, which is used in semantic segmentation tasks, with Atrous Spatial Pyramid Pooling (ASPP) to capture multi-scale contextual features [7]. The model is effective in identifying features of different sizes and shapes in landslide segmentation with enhanced accuracy of detection in complex terrain. DeepLabv3+ has an encoder-decoder architecture that ensures enhancements in boundary delineation due to the combination of high-level semantic and low-level spatial information. This leads to more accurate segmentation outputs as compared to the traditional models like FCN and U-Net. The atrous convolution is used to enable the model to enlarge its receptive field at a lower computational cost. Nevertheless, the model consumes a lot of labeled data to train effectively and this may pose a challenge in landslide research. Moreover, the model is not as applicable to real-time applications because of its computational complexity. The performance can also reduce in a non-homogenous environment where differentiating landslides and other features in the environment can be challenging. Further research can be aimed at streamlining the model to be more efficient and enhance its performance in situations with limited data. An attention-based architecture has become popular in enhancing the performance of segmentation in complicated image analysis problems.

Here, Attention U-Net is a variation of the standard U-Net model, which introduces an attention mechanism that allows the network to focus on the relevant spatial areas, but ignore the irrelevant background details [8]. This is especially significant in the landslide segmentation process where satellite images are usually characterized by non-homogenous terrain, vegetation, shadows, and noise which can adversely affect the model performance. The essence of Attention U-Net is the implementation of attention gates in the skip connections of the encoder-decoder.

These gates serve as filters which bring out key features concerning landslides and minimize the effects of the irrelevant areas. Consequently, the model enhances feature localization and improves segmentation accuracy without drastically boosting model depth. The architecture retains the benefits of U-Net, including effective reuse of features, and pixel-level classification accuracy, but mitigates its drawback of considering all features equally. Attention U-Net proved to be better than standard U-Net when used on landslide detection with data like Sentinel-2 photos and landslide catalogue maps. The model was more accurate and had fewer false positives especially in areas where there are landslides that have intricate backgrounds. It was also found to be more effective at detecting irregularly shaped landslides because it was more effective in focusing on salient features. Although there are these benefits, the model has some limitations. Its decreased ability to detect small-scale landslides is also one of the major issues as these landslides may not result in a robust feature response. Also, the attention mechanism adds new overhead to computation that increases the time of training and consumes more resources. Hyperparameter tuning is also sensitive to the performance of the model, which has to be optimized carefully depending on the dataset.

The further work in the field can be devoted to the efficiency of attention modules and integrating them with multi-scale or transformer-based strategies to make their performance even more efficient. Transformer-based models are relatively new models that have shown themselves as strong competitors to convolutional neural networks in semantic segmentation. SegFormer, a relatively small but powerful architecture that consists of a Mix Vision Transformer (MiT) encoder and a basic multilayer perceptron (MLP) decoder, is one of such [9]. Compared to the conventional CNN-based models, SegFormer can deal with long-range dependencies and global contextual information, which is essential in the correct localisation of landslides in large and complex remote sensing images. The MiT encoder captures features at varying scales in hierarchical features enabling the model to perceive fine details and the global patterns. This is especially useful in the case of landslides, where landslides may differ dramatically in size, shape, and texture. Instead, the MLP decoder makes the segmentation process more straightforward as it is able to combine the features efficiently without heavy computation being involved.

Its capability to record world relationships helps to better classify the landslides even in heterogeneous soils with different environmental settings. The model also relatively is efficient when compared to other transformer based methods, and thus is applicable in large scale applications. Nevertheless, there are limitations of SegFormer, despite its advantages.

Transformer-based models generally require large amounts of training data to achieve optimal performance, which can be a challenge in landslide studies where labeled datasets are limited. Also, the model requires considerable computational power especially when training, limiting its applicability in resource-limited settings. The quality and diversity of the training data is also a crucial factor in the performance of the model. Future studies can be aimed at enhancing the efficiency of data, decreasing the computing needs, and including hybrid structures that merge the merits of CNNs and transformers.

The current research [10] on landslide detection indicates that there is a drastic change in the conventional approaches to the development of the modern remote sensing and deep learning-based approaches. The methods used in the past mainly depended on manual interpretation, geomorphological analysis, and field surveys which were time consuming, labor intensive and could not be scaled to a larger spatial scale. As remote sensing technologies have developed, satellite imagery, aerial photography and LiDAR data have been widely used to enhance the precision of the detection. LiDAR, in particular, has been found especially useful in the high-resolution topography data collection and landslide detection even when covered with thick vegetation; nevertheless, it is expensive and not very accessible, which limits its application in large scale. Therefore, the use of optical remote sensors data and machine learning and deep learning methods has become a topic of growing interest among researchers.

Convolutional neural networks (CNNs), including their versions, e.g., Fully Convolutional Networks (FCNs), U-Net, and Mask R-CNN have demonstrated potential in recent years to automatically extract landslide features in complex terrains. These models are based on hierarchical feature learning that helps in increasing the accuracy of segmentation and classification. In spite of these developments, much of the research has mainly emphasized on the identification of recent or active landslides which have obvious physical appearances. Conversely, old or dormant landslides are a problematic issue to identify because of their faint geomorphic markers, and resemblance to adjacent landscapes. Also, large, well-annotated, and publicly available datasets are scarce, which is a significant constraint and often causes overfitting and poor generalizability of models. Recent studies highlight the combination of transfer learning, data augmentation, and multi-source data fusion to increase model robustness to overcome these challenges. Moreover, there is an increased desire to come up with generalized frameworks that can identify various forms of landslides such as ice avalanche and complex terrain failures, which would enhance the practical relevance of the frameworks in disaster management and monitoring systems of the environment.

2.3 SOFTWARE REQUIREMENT SPECIFICATION:

2.3.1 Purpose

The main purpose of this project is to create and develop a smart and automated landslide segmentation system based on the latest deep learning methods. One of the most destructive natural hazards that cause devastating damage to infrastructure, loss of human life and long-term environmental impairment are landslides. These occurrences tend to be unforeseen and influenced by numerous factors such as rainfall, geology, slope instability and land-use patterns. Therefore, the area with landslides should be identified and properly as soon as possible to minimize the risks and be able to react to the disaster. The project aims to use high-resolution satellite images in conjunction with the contemporary deep learning models to automatically detect and segment landslide areas. Satellite images give a detailed and extensive picture of the landscape and hence is an important source of tracking geographical changes with time. However, analysis of such data cannot be done manually due to its time-consuming and difficult nature due to complexity of the terrain features and variations in environment conditions. To eliminate this issue, the proposed system will be aimed at automating the process with data-driven models which are able to learn complex patterns and distinguish between landslide areas and non-landslide areas. Among the objectives of the work, there is the application and testing of the state-of-the-art neural network models, such as Attention U-Net that enhance the performance of segmentation by focusing on the areas of an image that are important in terms of the segmentation task performance. The attention mechanism enables the model to focus on the significant features, such as landslide boundaries and changes in the texture and ignore the insignificant background data. This selective feature learning enhances the model to locate landslide regions more accurately even in situations where the difference between affected and unaffected regions is small. In addition to improving the precision of the segmentation, the project will design the system that will be resilient and can be used in any kind of terrain and environmental conditions. Landslides may be experienced in different topographies such as mountainous, forested and those with different soil composition. As such, the model should be in a position to adjust to these changes and continue performing equally when used in new and unknown data. This level of generalization is critical to practices in the real world where it can be used in other geographical locations. The other important project objective is to reduce necessity of satellite images manual interpretation. The conventional systems of landslide mapping are usually based on a subjective approach to the analysis and may be time-consuming, labor-intensive and prone to human error. The proposed system would save a lot of effort needed

in the analysis process and enhance consistency and reliability by automating the segmentation process. Not only will this speed up the process of identification, but also allow more frequent monitoring of the areas prone to landslides. Decision making processes among the different stakeholders, such as the disaster management authorities, urban planners and environmental monitoring bodies, are also meant to be facilitated by the developed system. Planning of evacuation measures, less risky infrastructure designs, and preventive actions can be based on proper and timely data regarding the areas affected by landslides. In addition, automated outcomes of the segmentation should be provided, which will enhance situational awareness in case of an event of a disaster and allow the authorities to respond more effectively. The consideration in this project is also the issues of scalability and efficiency. The system will be capable of handling large volumes of the satellite data and generate outputs of segmentation in a manageable time frame. This is especially critical to real or near real-time applications, when quick analysis is necessary to react to an on-going or possible landslide. The project will also aim to trade-off between accuracy and the performance of the computation process by combining effective deep learning models. Lastly, this project will also be geared towards making a contribution towards the creation of a scalable, efficient and reliable landslide monitoring and risk assessment solution. The system integrates the most recent deep learning algorithms with satellite images to provide a strong solution to the issue of landslide area detection with high accuracy. Not only does this work highlight the potential of artificial intelligence in the geospatial domain, but also justifies the efforts to enhance disaster preparedness, mitigate risks, and safeguard human lives and the environment.

2.3.2 Scope

The project will cover the design and development of a deep learning-based system to perform the semantic segmentation of landslides using multi-source/ multi-temporal satellite images. The main emphasis is to develop an automated system that would analyze extensive geospatial data and derive significant spatial features that would reveal the occurrence of landslides. Due to the topography of the land and changes in the environment, the system is expected to offer a dependable and effective method of locating potential and actual landslides areas with high accuracy. The major feature of the project is the gathering and usage of appropriate datasets, namely the Landslide4Sense dataset, which comprises annotated satellite images and ground truth masks. With these annotations, it is possible to have supervised learning and the model is able to create a distinction between the landslide and non-landslide regions on the pixel level. The data is used to train, validate and test the segmentation models so that the system learns across the various terrain conditions and geological variations.

It also involves the comprehensive data preprocessing methods that can improve the quality and usability of the input data. These include steps like scaling images to a uniform size, standardizing pixel values to stabilize the training process, and using data augmentation like flipping, rotation, and scaling to make the dataset more diverse. Moreover, noise removal and quality inspection are done to remove the corrupted or inconsistent samples. Such preprocessing procedures are necessary in enhancing the robustness of the model and in providing better generalization in the various conditions of the environment. The most important aspect of the project is the introduction of the state-of-the-art deep learning architectures based on the models, namely Attention U-Net. The architecture builds on the traditional U-Net and introduces attention mechanisms that can make the model pay attention to the relevant spatial areas and ignore irrelevant background information. Such a selective feature learner improves the capability of the model to effectively identify landslide boundaries and complex terrain structures. The framework can also support other current architectures in addition to Attention U-Net, to achieve even better segmentation results.

A model training and evaluation process will also be part of the project, in which the performance of the segmentation model will be evaluated based on traditional evaluation metrics. These are Intersection over Union (IoU), Dice coefficient, precision, recall, and accuracy which when combined together allow a complete picture of the effectiveness of the model. These measures are used to determine the consistency of the predicted segmentation with the ground truth and to guarantee that the model is reliable in a variety of situations. The other significant part of the scope is the creation of a scalable and adaptable system. The framework is set to process large amount of satellite data and can be scaled to other geographical areas with diverse terrain features. It is an important adaptability because landslides take place in varying environments, and the system should be able to sustain similar performance in various conditions. The model is also scalable to allow it to be expanded to wider use, like landslide monitoring across a region or country.

The project also highlights automation in landslide monitoring, which will decrease manual satellite image analysis. The classical approaches are based on the high level of expertise analysis, which may be rather time-consuming and be subject to inconsistency. The system allows the quicker and more dependable identification of landslide zones by means of automating the segmentation process, which will facilitate near real-time analysis. This is especially beneficial in the situation of disaster management and emergency response, where prompt information is of the essence. Moreover, the framework is constructed in a manner that

is future extensible. It may be improved with the help of inclusion of more information sources including Digital Elevation Models (DEM) that offer the data concerning the terrain height and slope, and weather-related data including the amount of rainfall and soil moisture.

These other inputs will enhance the predictive abilities of the system, as it can not only detect but also provide early warning and risk estimation of the possible landslide occurrences. Altogether, the pipeline of this project encompasses the entire landslide segmentation process, including data collection and preprocessing, model development, evaluation, and deployment issues. It will provide a reliable, effective, and scalable solution applicable in practice to monitor landslides and manage disaster risks, as well as offer a solid base of further studies and improvements in this field.

2.3.3 Software Requirements

Operating System

- Windows 10/11 (64-bit)
- Linux-based environment (optionally scalable and deployable) compatible.

Programming Language

- Python 3.8 or above
- Preferred due to its extensive support for machine learning and data processing libraries

Frameworks and Libraries

PyTorch

Applicable to designing, training and evaluating deep learning models including Attention U-Net. It offers the ability to customize models and efficient use of GPUs.

OpenCV

Applied in image preprocessing application such as image resizing, image filtering, image augmentation, noise reduction, etc.tion. It aids in enhancing quality of image input to enhance performance of models.

NumPy-Pandas

Applied to perform efficient data manipulation, numerical calculations and data handling.

Matplotlib/Seaborn

Applied to visualize training output, loss curves, accuracy plots, and compare model performance.

Torchvision

Offers ready-to-use datasets, image manipulations, and model tools to facilitate deep learning piping.

Tools and Platforms

Kaggle

Applied in accessing data sets, training models, and experimentation. It offers GPU support and interactive development and testing environment.

Jupyter Notebook / Google Colab (optional)

Applied in experimentation and visualization, as well as step-by-step execution of code.

Hardware Requirements (Recommended)

- Processor: Intel i5 or higher
- RAM: Minimum 8 GB (16 GB recommended for large datasets)
- GPU: NVIDIA GPU with CUDA support (for faster model training)
- Storage: Minimum 10–20 GB free space for datasets and models

Functional Requirements

- The system must be able to receive satellite pictures as input.
- It must preprocess images to reduce noise and normalize images..
- Syntactic segmentation will be done by the model to detect landslide areas..
- The system must also produce fragmented images with areas of concern.
- It must measure performance in terms of IoU, Dice score and accuracy..

Non-Functional Requirements

- **Performance:** The system must be capable of giving the correct segmentation results with reasonable processing time.
- **Scalability:** It must be able to manage large amounts of data and be flexible to other areas.
- **Reliability:** model must give similar results to different inputs.
- **Usability:** The system is to be simple to use and understand by the users with a basic technical knowledge.

CHAPTER-3
PROPOSED METHOD

3.1 DESIGN METHODOLOGY:

1. Data Collection and preprocessing

The data used in the present research was Landslide4Sense which served as the main source of the data to be applied in the development and testing of the landslide segmentation model. The dataset is tailored to the task of landslide detection and is a collection of patches of high-resolution satellite images and their pixel-wise annotated masks. The masks are binary representations whereby the single pixels are classified as landslide (foreground) or non-landslide (background), which allows supervised learning to be done accurately in the segmentation task. The data is systematically categorized into three different sets, which include training, validation, and testing.

The subsets have distinct folders of input images and ground truth masks. The model is trained on the training set to learn spatial patterns, textures and distinguishable features of landslides. The validation set is applied in the training process to keep track of the model performance, hyperparameter fine-tuning, and overfitting. Lastly, the test set will be used to test the generalization ability of the trained model on unknown data, which will provide a fair evaluation of performance. A number of preprocessing procedures were used before the data was fed into the deep learning model to promote consistency and the efficiency of the model. Originally, all images and masks were scaled to the same size to ensure that they were compatible with the network architecture.

This is necessary as deep learning models need inputs that have fixed sizes. The pixel values of the images were then normalized (usually to a range of 0-1) which facilitates stabilization and speeds up the training process. To supplement resizing and normalization, close alignment between images and associated masks was checked to make sure that an input image is properly matched with the ground truth annotation.

To preserve data quality, any mismatched or corrupted samples were discarded. Moreover, simple data enhancement methods, including horizontal flipping, vertical flipping and rotation, can be attempted to enhance variety of the dataset, and the resilience of the model to changes in terrain and environmental conditions. In general, the preprocessing step is vital in making the data ready to be trained successfully in the model. These measures can help to enhance

segmentation accuracy and generalization performance of landslide detection tasks by means of uniformity, data quality and variability.

2. Data Splitting

The dataset (Landslide4Sense) utilized in this paper was properly separated into three parts that included training, validation, and testing to provide a systematic and credible model development. Correct data splitting is an important step in machine learning workflow since it directly affects the learning capability of the model and its generalization capability to unknown data. The training set was assigned to 70 percent of the dataset, which is used in learning a model. In this step, the model is introduced to input satellite images together with their respective ground truth masks and it can learn intricate spatial patterns, textures, and boundaries related to landslide-prone areas.

The proportion of training data is relatively higher so that the model will capture various terrain characteristics and differences in the environmental conditions which is critical in development of a robust segmentation system. Validation set is a subset of the dataset (20 percent) that is very important in the training process. This subset is also not utilized to re-estimate model weights as the training set but is rather a performance monitoring device. The model is then tested on the validation data at the end of every training iteration or epoch to determine its learning performance. This can be used to detect overfitting, where the model fares well on training data, but poorly on unseen data.

Through monitoring validation metrics (e.g., loss and accuracy) suitable changes to hyperparameters, like learning rate, batch size and number of epochs, can be made. This repeated process of tuning the model makes sure that the model is in a balance between learning and generalization. The testing set was set aside to comprise 10% of the dataset and is only used to assess the final model. This subset is not visible in any way during training and validation, thus it is an unbiased parameter to evaluate the actual model performance. Performance on the test set is used to give a realistic idea of the performance of the model in the real world when applied to new satellite images. The measures received at this step, e.g., segmentation accuracy or overlap-based ones, indicate how well the trained model can identify landslide areas. To guarantee the integrity of data, care was taken to make sure that there was no overlap of the three subsets.

All images and all the corresponding masks were contained in a single subset, excluding data leakage and having a fair evaluation process. Moreover, the sample distribution of the landslides and non-landslides was maintained uniform in all the subsets, in order to prevent

bias and to make sure that each of the subsets portrays the general characteristics of the data set. This 70:20:10 splitting strategy offers a reasonable balance of data to base on training and still retain enough data to validate and test. It allows keeping track of performance continuously, facilitates efficient hyperparameter tuning, and provides an objective analysis of the model. In general, this systematic data fragmentation approach will play an important role in the development of a robust and generalizable landslide detection system.

3. Model Architecture:

The deep learning architectures used in this project are SegFormer and MobileViT, which are advanced models based on transformers and hybrid models that are designed to enhance the accuracy of segmentation in complex and heterogeneous terrain. The choice of these models was aimed at addressing the gaps of conventional convolutional neural network (CNN)-based models, especially, their inability to learn long-range correlations and global contextual features. Detection of landslides using satellite imagery is not an easy undertaking because of the differences in land topography, vegetation cover, lighting and the dissimilarities in the shapes of the landslides. As such, models that are capable of acquiring local spatial and global contextual relationships, which are key to proper segmentation, must be employed.

SegFormer is a transformer-based semantic segmentation network that proposes a new way of doing things by instead of using the traditional convolutional encoder, a hierarchical transformer encoder is used. The model starts by breaking down the input satellite image into smaller patches via a patch embedding technique. The patches are then converted into feature representations and run through multi-head self-attention mechanisms which allow the model to learn about long-range connections between remote parts of the image. This can be especially helpful in landslide detection where the features of interest can be distributed over extensive regions and demand international knowledge. One of the strengths of SegFormer is that it provides the multi-scale feature representations. This hierarchical encoder gets the features at various resolutions and the model is able to extract fine features, like the boundaries and the texture, and high-level semantic features, like the terrain structure.

SegFormer does not heavily use positional encoding, unlike most transformer-based models. Rather, it applies a design that naturally maintains spatial data, and thus is more flexible and adaptable to images of different sizes and resolutions, which is particularly advantageous when dealing with satellite data. SegFormer has a decoder component that is implemented to be lightweight and efficient. It takes the multi-scale features produced by the encoder and fuses them using simple multilayer perceptron (MLP) layers to generate the final segmentation

output. This decoder works well and is simple; it combines information at various levels of features leading to accurate pixel predictions. A strong transformer encoder and an efficient decoder allow SegFormer to perform highly yet remain computationally efficient and thus is well adapted to large-scale geospatial problems. Conversely, MobileViT is a hybrid model that integrates the advantages of convolutional neural networks and vision transformers into a single model.

It is a model aimed at balancing performance and efficiency, which makes it applicable both to high accuracy tasks and can be used in resource constrained environments. The convolutional layers in MobileViT extract the local features, e.g., edges, textures, and fine-grained spatial details. Such characteristics are critical in disrupting small changes in the terrain, as well as subtle patterns that can be linked to landslide areas. MobileViT also uses lightweight transformer blocks to improve this local feature extraction, extracting global contextual information. These transformer modules are presented with feature representations which are modeled by capturing interrelationships between various parts of the image which allow the model to capture large scale spatial relationships in the image.

MobileViT is able to learn detail and contextual information using the same architecture, by combining convolutional operations with transformer-based attention mechanisms. Another important advantage of MobileViT is its computational efficiency. The model has fewer parameters and optimized operations, which makes it a lot lighter than all-transformer-based architectures. This efficiency enables it to be applicable in real-time and on any device with a limited amount of computational resources, without substantial performance impact.

Although it is lightweight, MobileViT still provides good segmentation quality as both local texture and global structures are well-captured. In general, SegFormer combined with MobileViT offers an excellent and holistic solution to landslide classification in this project. SegFormer is very efficient in capturing global context and multi-scale features and therefore can be very useful in understanding the large and complicated terrain variations. MobileViT on the other hand provides a middle ground by being local and global feature learners with high computational efficiency.

These architectures combine to improve the performance of the model in terms of precise identification of the landslide areas with complex boundaries and varying environmental features. Their complementary advantages lead to better segmentation, better generalization and better applicability to real world landslide monitoring and risk assessment system.

4. Model Evaluation:

After training the segmentation models, their performance was evaluated on the test dataset to assess how accurately they could identify landslide-affected pixels. Each model outputs a binary segmentation mask, where each pixel is classified as either *landslide* or *non-landslide* based on the predicted probability. The following performance metrics were used:

Accuracy: Measures the overall proportion of correctly classified pixels.

Precision: Indicates how many of the pixels predicted as landslide were actually correct (useful for reducing false positives).

Recall: Measures how many actual landslide pixels were correctly identified (useful for minimizing missed landslides).

F1-Score: Harmonic mean of precision and recall, providing a balanced metric for imbalanced datasets.

Intersection over Union (IoU): Evaluates how well the predicted landslide region overlaps with the ground-truth segmentation mask.

Dice Coefficient: Measures the similarity between predicted and actual masks and is widely used in medical and remote sensing segmentation tasks.

Confusion Matrix: Provides a pixel-level view of true positives, true negatives, false positives, and false negatives, helping analyze model mistakes.

3.2 SYSTEM ARCHITECTURE:

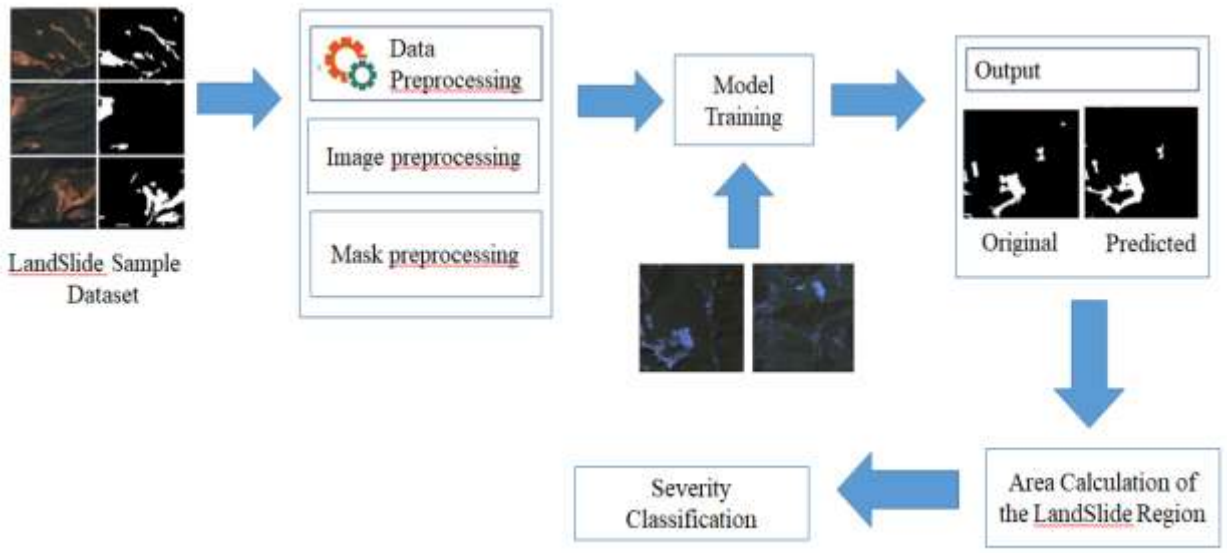

Figure 3.2.1: Architecture Diagram

Fig. 3.2.1 is a representation of the general structure of the proposed landslide segmentation system, which combines deep learning models with post-processing tools to offer an end-to-end automated system in detecting and assessing the risk of landslides. The workflow starts with Landslide4Sense dataset which is a collection of high-resolution satellite images accompanied with ground truth masks which are regions of landslides. These images are complex terrain conditions such as vegetation, shadows, and different light conditions and proper segmentation is not an easy task. The main input used to train and evaluate the deep learning models is the dataset. The second step is a data preprocessing step which is vital in enhancing the performance of the model and in maintaining uniformity in the input throughout the system. In this stage, all images are rescaled to constant size of 256256 pixels to ensure consistency and minimize the complexity of calculations. The value of pixels of the images is brought to a scale of [0,1], and this is useful in stabilizing the training process and speeding up convergence. In the same manner, the ground truth masks are also preprocessed by converting them to binary format with landslide areas denoted by the 1 and non-landslide areas denoted by 0. This preprocessing step is used to make sure that the models are capable of learning the difference between the affected and unaffected parts. The data is also preprocessed and then undergoes the model training phase, which involves using two significant deep learning models, SegFormer and MobileViT. The models are chosen because they are able to represent both the contextual and the spatial information. SegFormer is a transformer-based segmentation network that uses a hierarchical encoder to get multi-scale information of the input images. It employs self-attention to learn long-range dependencies and global contextual information, which comes in handy especially when it comes to comprehending large-scale terrain patterns in landslide detection. Its lightweight decoder also effectively combines multi-scale features, leading to pixel-wise accurate segmentation.

The model is made up of four key steps which include Dataset Input, Preprocessing, Model Training and Prediction Output. The flow diagram illustrates the flow of the input satellite images through each step, until the final segmentation masks are obtained.

1. Landslide Sample Dataset

The first step of the pipeline is the Landslide4Sense dataset that consists of paired satellite RGB images and binary masks that point to landslide and non-landslide areas. These are the sample images, which are the raw input to the system.

2. Data Preprocessing

Both images and masks are preprocessed before they are fed to the model:

Image Preprocessing:

All RGB images are downsampled to 256 256 pixels and brought to the range [0,1] in order to have a uniform input and reliable model training.

Mask Preprocessing:

Individual binary masks are scaled to 256×256 pixels and converted to single-channel. The values are normalized to [0,1], with 1 being pixels of landslides and 0 being background.

This preprocessing step ensures image–mask alignment and standardized format for the Models.

3. Model Building

U-Net Model:

U-Net is an encoder-decoder convolutional neural network to perform semantic segmentation.

This research employs it as the baseline to segment landslides. The input in the model is a 256 x 256 x 3 satellite image.

Features are acquired by the encoder (contracting path) through convolutional layers and scaling down spatial dimensions and scaling up depth.

A bottleneck layer forms high level contextual information with richer features representations.

The decoder (expanding path) uses transposed convolutions to upsample the data to recover spatial resolution.

Skip connections are made between encoder and decoder layers in order to maintain spatial information.

The final result is produced by applying a 1x1 convolution with sigmoid activation to give a binary segmentation mask.

Training and optimization of the model occurs via binary cross-entropy loss and Adam optimizer. Dice coefficient and IoU score are used to measure performance.

Deep Deconv ResU-Net:

DDR U-Net is a more advanced version of U-Net, which adds residual connections to improve the learning of features.

It processes 256 × 256 × 3 satellite images using an encoder–decoder structure.

The encoder employs convolutional layers and residual blocks, batch normalization and ReLU activation.

Identity connections (residual) assist in alleviating vanishing gradient issues and enhance stability in training.

The bottleneck layer attends to the rich spatial and contextual features as the filter depth increases. The decoder does the upsampling by transposed convolutions and synthesizes encoder features. More convolution blocks of residual convolution sharpen the reconstructed feature maps.

The last layer has a 1x1 convolution and is followed by a sigmoid activation that produces a binary segmentation mask.

This model offers an enhanced representation of features and accuracy in boundaries as compared to conventional U-Net.

Attention U-Net Model:

Attention U-Net is an improvement of U-Net that incorporates attention to emphasize the important parts. It is fed with a 256 x 256 x 3 satellite image to segment landslides.

The encoder is based on the normal U-Net architecture with convolutional and pooling layers.

In skip connections, attention gates are added in order to select significant features. The attention mechanism is directed by a gating signal of the decoder.

The attention mechanism inhibits the background irrelevant information and focuses on landslide areas.

The decoder upsampling features maps and fuses them with attention-refined encoder features. Additional convolutional layers sharpen the result of the segmentation. A 1 x1 convolution with sigmoid activation is used to obtain the final output.

Training of the model is based on binary cross-entropy loss and Adam optimizer.

It enhances the performance of segmentation particularly in arduous terrains where boundaries are not well defined.

SegFormer Model:

SegFormer is a semantic segmentation model that is a transformer that is structured to effectively retrieve local and global contextual information.

This study uses SegFormer to enhance the quality of landslide segmentation in rugged terrain. A 256 x 256 x 3 satellite image is the input to the model.

The hierarchical transformer encoder is used to extract features, with patch embedding to reduce the spatial dimensions and long-range dependencies with self-attention.

Multi-scale feature representations generated by the encoder can be used to comprehend terrain variability and patterns of landslides.

A bottleneck layer is deep contextual information that has rich semantic characteristics. The decoder is light and fuses features and upsamples to recover the spatial resolution.

In contrast to the traditional models, SegFormer is an efficient multi-scale feature combining model that does not require expensive computations.

The result is then obtained after a 1x1 convolution with a sigmoid activation to obtain a binary segmentation mask.

The model is trained on binary cross-entropy loss and is optimised by Adam optimiser.

The performance is measured in terms of Dice coefficient and IoU score which has better global feature learning and accuracy of segmentation.

MobileViT Model:

MobileViT is a model of hybrid deep learning, which is a combination of convolutional neural networks and transformer based attention.

In this study, MobileViT is employed to realize an effective and precise landslide segmentation. This model accepts a satellite image of 256 x256 x 3.

Convolutional layers are used to extract the initial features and capture local spatial information (edges and textures).

The feature maps are then transformed into sequences through MobileViT blocks and self-attention is used to learn the global relationships.

This mixture enables the model to acquire small-scale and high-level dependencies both.

This is because a bottleneck layer entraps high-level semantic properties of local and global information.

The decoder uses transposed convolutions to upsample and encoder features through skip connections.

Further convolutional layers are used to refine the segmentation output to enhance boundary detection.

A 1x1 convolution followed by sigmoid activation is used to produce the final output as a binary segmentation mask.

Training of the model is based on binary cross-entropy loss and Adam optimizer.

MobileViT is a tradeoff between performance and computational efficiency, which is appropriate to real-time applications.

4. Model Training

The model is trained on the preprocessed images and masks. The Attention U-Net is trained with binary cross-entropy loss and an Adam optimizer to learn how to recognize patterns of landslides. Training enhances the model to distinguish the landslide areas as compared to the terrain.

5. Output: Original vs. Predicted Mask

After the model is trained, the model gives predictions of landslides on unknown test images: Original Image: The original image of the dataset. Predicted Mask: The model results in a segmentation map. It is evident that the model recognizes landslide areas by making the locations of detected areas visible in the predicted binary mask.

3.3 DESCRIPTION OF ALGORITHM

Algorithm: U-Net for Landslide Segmentation

Input: Landslide4Sense dataset, resized to 256×256×3

Output: Segmentation mask (256256), each pixel is considered to be landslide or non-landslide.

Method:

1. Initialize U-Net Architecture
2. Input Image
   - Shape : 256×256×3
3. Construct U-Net Model
   - Encoder (Contracting Path)
     - Repeated blocks of Conv2D → ReLU → MaxPooling
   - Bottleneck Layer
     - Two Conv2D layers with increased filters
   - Decoder (Expanding Path with Attention)
     - Upsampling via Conv2DTranspose
     - Concatenation with corresponding encoder feature maps (skip connections)
     - Conv2D layers to sharpen result.
   - Final Result Layer
     - Upsampling via Conv2DTranspose
     - Skip connections (concatenation with respective encoder feature maps)
     - Conv2D layers to smooth output
4. Train the Model
   - Loss: Binary Crossentropy
   - Optimizer: Adam
   - Monitor: Dice coefficient, IoU and validation loss
5. Return the Trained U-Net Model.

4. Train the Model

- Loss: Binary Crossentropy
- Optimizer: Adam
- Monitor: Dice coefficient, IoU and validation loss

Algorithm: DeepDeconvResU-Net for Landslide Segmentation

Input: Satellite image (256×256×3)

Output: Binary mask (256×256) indicating landslide regions

Method:

1. Initialize DeepDeconvResUNet Model
2. Input Image
   - Shape: 256×256×3
3. Construct DeepDeconvResU-Net Architecture
   - Encoder (Contracting Path)
     - Several residual blocks: each containing Conv2D → BatchNorm → ReLU → Add skip
     - Downsampling using MaxPooling. Down-samples spatial dimensions, and allows the model to build up wider contextual information on more than one scale.
   - Bottleneck Layer
     - Residual block with maximum filters
     - Represents rich and abstract feature representations, combining global context with learned features from the encoder.
   - Symmetric Decoder Path
     - Upsampling via Conv2DTranspose. Reconstructs spatial resolution progressively, by rescaling feature maps.
     - Block of residue at each upsampling. Enhance the features that are being upsampled and refine them to increase the accuracy of segmentation by preserving the essential information.
     - Omit encoder to decoder connections. Combine high-resolution encoder properties with decoder properties to retrieve fine spatial properties and enhance localization.
   - Final Output Layer

- 1×1 Conv2D using sigmoid activation to predict binary segmentation mask. Converts transform maps into pixel-wise likelihoods, indicating whether the pixel is a landslide or not.

4. Train the Model
   - Loss: (Binary) Cross-Entropy.
   - Optimizer: Adam (learning rate 1e-4).
   - Train on mini-batches from the DataGen; validate on other validation set.
   - Keep an eye on the training/validation accuracy, loss, and pixel-wise (Precision, Recall, F1, IoU, Dice).
   -

4. Return the Trained DeepDeconvResUNetUNet Model.

Algorithm: Attention U-Net for Landslide Segmentation

Input: Satellite image (256×256×3)

Output: Binary mask (256×256) indicating landslide regions

Method:

1. Initialize Attention U-Net Model
2. Input Image
   - Shape: 256×256×3
3. Construct Attention U-Net Architecture
   - Encoder (Contracting Path)
     - Repeated block of: Conv2D → ReLU → Conv2D → ReLU → MaxPooling
     - Feature maps saved at each level for skip connections (c1, c2, c3, c4)
   - Bottleneck Layer
     - Two Conv2D layers with 1024 filters and ReLU activation.
   - Decoder (Expanding Path with Attention)
     - For each decoder level Generate a gating signal from the deeper layer using 1×1 Conv2D.
     - Apply an Attention Gate on the corresponding encoder feature map:
       - Project encoder feature and gating signal → add → ReLU → 1×1 Conv2D → sigmoid
       - Multiply attention map with encoder feature to suppress irrelevant regions.

        - Upsample decoder feature using Conv2DTranspose.
        - Concatenate upsampled feature with attention-filtered skip feature.
        - Apply two Conv2D + ReLU layers to refine the fused features.

4. Train the Model
    - Loss: Binary Cross-Entropy.
    - Optimizer: Adam (learning rate 1e-4).
    - Train on mini-batches from the DataGen; validate on a separate validation set.
    - Monitor training/validation accuracy, loss, and pixel-wise metrics (Precision, Recall, F1, IoU, Dice).

4. Return the Trained Attention UNet Model.

Algorithm: SegFormer for Landslide Segmentation

Input: Satellite image (256×256×3)

Output: Binary mask (256×256) of landslide regions

Method:

1.The first step is to initialize Attention U-Net Model.

2. Input Image
    - Shape: 256×256×3

3. Construct SegFormer Architecture
    - Encoder
    - Patch Embedding using Conv2D (stride = 4) to reduce spatial size
    - Convert feature maps into sequence (flattening)
    - Go through multiple Transformer Blocks
        - Multi-head self-attention captures global dependencies
        - Feed-forward layers refine feature reprsentations.
    - Bottleneck Layer
        - Deep transformer model of the global contextual information.
        - Codes associations among remote areas (significant to landslides)
    - Decoder
        - Apply Projection to encoder features
        - Perform **Upsampling** to restore spatial resolution
        - Multiscale (between encoder stages) features.
        - Combine features using simple MLP layers.

- Output Layer
    - Apply 1×1 convoloution.
    - Use sigmoid activation
    - Produce final binary segmentation mask (256×256×1)

4. Train the Model
    - Loss: Binary Cross-Entropy.
    - Optimizer: Adam (learning rate 1e-4).
    - Train using mini-batches from the DataGen; validate on a separate validation set.
    - Track the accuracy of training/validation, loss, and pixel-level metrics (Precision, Recall, F1, IoU, Dice).

4. Return the Trained SegFormer Model.

Algorithm: MobileViT for Landslide Segmentation

Input: Satellite image (256×256×3)

Output: Binary mask (256×256) indicating landslide areas.

Method:

1. Initialize MobileViT Model
2. Input Image
    - Shape: 256×256×3
3. Construct MobileViT Architecture
    - Encoder
    - Apply repeated blocks of : Conv2D → ReLU → Conv2D → ReLU → MaxPooling
    - Extract hierarchical spatial features at multiple levels
    - Store feature maps for skip connections
    - MobileViT Block
        - Using MobileViT block at bottleneck layer.
        - Local feature extraction: convolution..
        - Convert feature maps into sequence (flattening).
        - Using lightweight multi-head self-attention, extract global dependencies (image-wide).
        - Use refining layers by application of feed-forward.
        - Reconstruct sequence back to spatial feature maps
    - Bottleneck Layer

        - Deep feature representation combining CNN and transformer features.
        - Codes local textures and global context information of landslides.
    - Decoder
        - Up-sample with Conv2DTranspose.
        - Join with respective encoder characteristics (skip connections)
        - Restore spatial resolution progressively
    - Output Layer
        - Apply 1×1 convoloution.
        - Use sigmoid activation
        - Generate final binary segmentation mask (256×256×1)

4. Train the Model
    - Loss: Binary Cross-Entropy.
    - Optimizer: Adam (learning rate 1e-4).
    - Train on mini-batches from the DataGen; validate on a separate validation set.
    - Monitor training/validation accuracy, loss, and pixel-wise metrics (Precision, Recall, F1, IoU, Dice).

4. Bring back The MobileViT Trained Model.

3.4 DATASET DESCRIPTION AND REQUIREMENT SPECIFICATION

3.4.1 Dataset Description

The dataset to be employed in this project is the Landslide4Sense dataset which is free to access on Kaggle, and is commonly used in the research of landslide segmentation. It comprises high-resolution satellite images of the areas before and after landslides, and associated ground truth binary masks that are quite obvious where landslides took place. The images are labeled pixel-by-pixel, with the landslides clearly delineated by the background, making it possible to use these images in semantic segmentation. The dataset is specifically aimed at deep learning usage and is a stable benchmark to assess the effectiveness of segmentation models in disaster management and environmental monitoring.

The dataset consists of a large diversity of terrain features and geographical conditions, including mountainous areas, forested ones, rocky areas, and slopes of various heights. It also reflects fluctuations in weather conditions, seasons, growth of vegetation, and surface features. This variety allows the dataset to be very realistic and makes the model training process more

complicated since landslides may differ greatly in shape, size, and appearance in different places.

Several data preprocessing methods were used to ensure that the training is effective and that the model performs better. All pictures and the respective masks were scaled to even size of 256x256 pixels, to save uniformity and minimise complexity in calculations. The images were scaled to a normalized range (0 to 1) stabilizing the training process and enhancing the rate of convergence. The binary masks were also computed to make sure that there is a clear difference between landslide (1) and non-landslide (0) areas. Also, the artificial increase of the dataset diversity with data augmentation methods, including rotation, horizontal and vertical flipping, zooming and brightness adjustments, was employed to decrease overfitting.

The data was systematically divided into training, validation, and test subsets, which made it possible to evaluate the model performance appropriately. It trains the set of features representations, uses the validation set to perform hyperparameter tuning and avoid overfitting, and uses the test set to evaluate the final performance.

This well-organized system provides fair evaluation and enhances the validity of findings in experiments. In general, the Landslide4Sense dataset is an extensive and rich dataset of satellite images that can be used to create effective models of deep learning. Its pixel-wise annotations, multi-terrain coverage, and support of advanced architectures, including SegFormer and MobileViT, make it very appropriate to construct precise and scalable landslide detection systems.

3.4.2 Requirement Specification

Software Requirements:

Programming Language: Python 3.7+

Development Environment : Google Colab / Kaggle Notebooks

Frameworks and Libraries:

- TensorFlow / Keras – for model building and training
- NumPy, Pandas – for data manipulation OpenCV / PIL – for image processing
- Matplotlib, Seaborn – for result visualization
- scikit-learn – for evaluation metrics (accuracy, precision, IoU, etc.)

Hardware Requirements:

Processor: Intel i5 or higher

RAM: Minimum 8 GB

Graphics Card (GPU)

CHAPTER-4

RESULTS AND OBSERVATIONS

4.1 DESCRIPTION OF RESULTS

Confusion matrix: Confusion Matrix is a key performance measurement for the classification task in landslide segmentation.

It visually shows the correct and incorrect predictions of the model by comparing actual vs. predicted labels.

By analyzing the confusion matrix, you can observe how well the model distinguishes between landslide and non-landslide areas.

Performance Metrics:

Accuracy: Accuracy is the ratio of correctly predicted observations (both positive and negative) to the total observations. It is a measure of how often the model is correct overall.

$$Accuracy = \frac{True\ positives + True\ Negatives}{Total\ Instances} \quad \text{-(Eq 1)}$$

Precision: Precision is the ratio of correctly predicted positive observations to the total predicted positives. It is a measure of the accuracy of positive predictions.

$$Precision = \frac{True\ positive}{true\ positives + false\ positives} \quad \text{-(Eq 2)}$$

Recall: Recall is the ratio of correctly predicted positive observations to all observations in the actual class. It measures how well the model can identify positive instances.

$$Recall = \frac{true\ positives}{True\ positives + False\ negatives} \quad \text{-(Eq 3)}$$

F1 score: F1-Score is the harmonic mean of Precision and Recall.

$$F1\text{-}score = 2 * \frac{Precision * Recall}{Precision + Recall} \quad \text{-(Eq 4)}$$

IoU Score (Intersection over Union): IoU Score measures the overlap between the predicted positive area and the actual positive area, divided by the union of both the areas so it can also known as Intersection over Union It evaluates how well the predicted segmentation matches the ground truth.

$$IOU = \frac{\text{True Positive}}{\text{True Positive+False Postive+False Negative}} \quad \text{-(Eq 5)}$$

Dice Coefficient: The Dice Coefficient is a measure of overlap between two samples. It calculates the similarity between the predicted and actual areas by considering both precision

and recall. It is also known as the F1 Score for segmentation tasks.

$$Dice\ Coefficient = \frac{2*True\ \ Positive}{2*(True\ Positive+False\ Postive+False\ Negative)} \quad -(Eq\ 6)$$

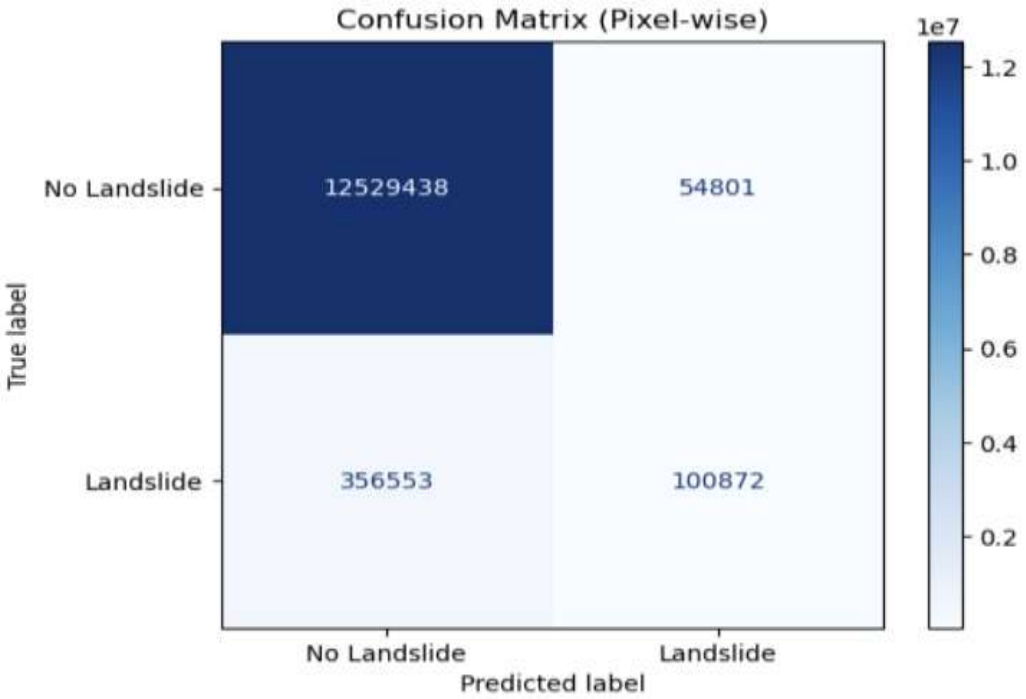

Figure-4.1.1: Confusion matrix of U-Net

Figure 4.1.1 displays the confusion matrix of the U-Net model illustrates a high precision of the model in identifying the areas of the U-Net model that can be classified as No Landslide, and 12,529,438 pixels were identified as true negatives. This shows that the model is very useful in locating the background areas, and reducing the false positives. Nonetheless, this model exhibits weaknesses on the detection of "Landslide" pixels as indicated by 356,553 false negatives, that is, genuine landslide areas have been categorized as non-landslide. This will lead to a relatively low recall of the landslide class meaning that a lot of landslide pixels will be overlooked during prediction. The total accuracy may seem high, given the prevalence of background pixels, but this may be false in disproportionate datasets, where one of the classes is much larger than the other. The unbalanced pixels between landslide and non-landslide influence the sensitivity of the model to the minority class. Thus, the detection of landslide pixels should be enhanced to increase the applicability of the model into the real-life scenario, particularly in disaster monitoring systems where missing crucial areas can be devastating.

```
📊 Classification Report:
              precision    recall  f1-score   support

No Landslide       0.97      1.00      0.98  12584239
   Landslide       0.65      0.22      0.33    457425

    accuracy                           0.97  13041664
   macro avg       0.81      0.61      0.66  13041664
weighted avg       0.96      0.97      0.96  13041664
```

Figure-4.1.2: Performance Metrics of U-Net

Figure 4.1.2 presents a classification report in detail, quantitatively evaluates the segmentation performance of the model by precision, recall and the F1-score of each of the classes. The No Landslide class has outstanding performance with a high precision (0.9846), a high recall (0.9859), and F1-score (0.9852) indicating a high ability of the model to classify and identify background pixels correctly. Conversely, the Landslide category reports an average performance, having a precision of 0.5970, a recall of 0.5757, and an F1-score of 0.5861, thus, meaning that the model is not very good at correctly identifying the pixels of a landslide owing to other factors like class imbalance, texture changes, and subtle changes in boundaries. With such a significant impact of the prevalence of background pixels, the overall accuracy of 97.15% is not representative of the actual performance of the model, whereas the macro-average F1-score of 0.7857 offers a more balanced assessment. These findings indicate that the model is very dependable in prediction of the background but more work is required to enable accurate localization of the landslides.

Figure-4.1.3: Confusion matrix of DeepDeconvRestU-Net

Figure 4.1.3 shows the classification report of the model, showing its performance in both classes No Landslide and Landslide. The overall accuracy of the model is high (approximately 97%), which means that most pixels are accurately assigned. In the case of the No Landslide class, the model is doing particularly well with a precision of 0.98, recall of 0.99 and F1-score of 0.99, indicating its high-level ability to detect background areas with a minimum amount of errors. In this case, or the "Landslide" class, the model has a precision of 0.78, meaning that when the model predicts a landslide, it is accurate most of the time. But the recall is relatively lower 0.57, implying that there are not all landslide pixels identified. F1-score of 0.66 indicates a trade-off between precision and recall, which means that it performs fairly well on the minority class. The macro and weighted averages also confirm that the model ensures that overall performance is stable even in the case of imbalance in the number of classes. The findings underscore this model to be consistent in prediction, although it can be enhanced to cover all landslide areas.

Classification Report:

	precision	recall	f1-score	support
No Landslide	0.98	0.99	0.99	11989824
Landslide	0.78	0.57	0.66	593088
accuracy			0.97	12582912
macro avg	0.88	0.78	0.82	12582912
weighted avg	0.97	0.97	0.97	12582912

Figure-4.1.4: Performance Metrics of DeepDeconvResU-Net

Figure 4.1.4 shows the confusion matrix that gives a detailed picture of the pixel-wise classification performance of the model. Using the model, there are 11,895,582 pixels of No Landslide correctly identified by the model, and the capability of the model to identify non-landslide area is high. It also rightly classifies 340, 258 landslide pixels, which proves its capability of identifying real landslide areas. There are however 252,830 false negatives (Landslide pixels are classified as non-landslide) which is the reason why the recall is lower in the classification report. There are also 94,242 false positives, i.e., the non-landslide regions are classified as landslides. Nevertheless, the amount of true positives and true negatives is far much more than the number of errors, and the overall accuracy is high. The confusion matrix also shows the effects of the class disparity where the prevalence of the background pixels

affects the performance parameters of the model. To ensure a better model application in real-world scenarios like disaster monitoring and early warning systems, better detection of landslide pixels, especially false negatives, would have improved the model.

Figure 4.1.5: Confusion Matrix of AttentionU-Net

Figure 4.1.5 demonstrates the confusion matrix of the Attention U-Net model, which can be used to determine the effectiveness of the model to identify the landslide and non-landslide areas per pixel. The diagonal components reflect accurate predictions where the model was able to classify 12,406,464 pixels as No Landslide and 263,317 pixels as Landslide meaning that the model was performing well in classifying both types. This shows that the Attention U-Net architecture has the ability to store spatial features and concentrate on the important regions by use of attention. The model works especially well in identifying the background areas because of the prevalence of non-landslide pixels in the data. But the off-diagonal elements show misclassifications that impact on overall performance. In particular, 177,775 non-landslide pixels were wrongly labeled as landslide (false positives) and 194,108 landslide pixels were incorrectly labeled as non-landslide (false negatives). These inaccuracies occur because of the irregularity and complexity of the landslide areas, and the disproportion of the classes. As the confusion matrix shows, the model is very accurate with most of the classes, but it is not very good at identifying smaller and less prominent landslides. The model could be improved by making it more sensitive to landslide pixels to make it effective in disaster monitoring in the real world.

```
Classification Report:
              precision    recall  f1-score   support

No Landslide     0.9846    0.9859    0.9852  12584239
   Landslide     0.5970    0.5757    0.5861    457425

    accuracy                         0.9715  13041664
   macro avg     0.7908    0.7808    0.7857  13041664
weighted avg     0.9710    0.9715    0.9712  13041664
```

Figure 4.1.6: Performance Metrics of AttentionU-Net

Figure 4.1.6 shows the classification report, which is a quantitative assessment of the Attention U-Net model in terms of precision, recall, and F1-score. No Landslide class has a high performance, with precision of 0.9846, recall of 0.9859, and F1-score of 0.9852, which means that it correctly detected the background pixels with a very small error. The encoder-decoder structure and attention gates are attributed to this strong performance because they preserve spatial information, and suppress irrelevant features. The overall accuracy of 97.15% further ascertains that the model has strong general performance.Conversely, the Landslide category also records moderate results, meaning that it has a precision of 0.5970, a recall of 0.5757 and an F1-score of 0.5861, resulting in the fact that detection of landslide areas is not a simple task. The reduced recall indicates that some of the landslide pixels are not detected, which is in line with the false negatives recorded in the confusion matrix. A more objective assessment is the macro-average F1-score of 0.7857, which shows the effect of the imbalance of classes on the performance of the model.

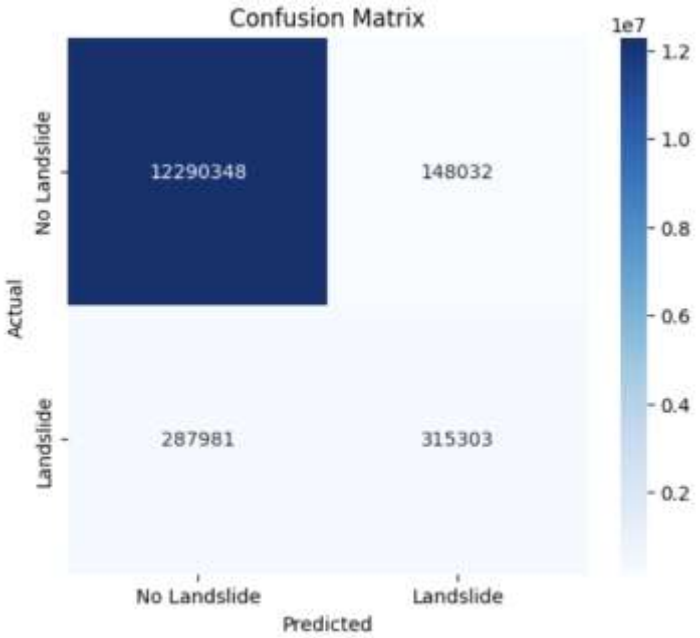

Figure 4.1.7: Confusion Matrix of SegFormer

The confusion matrix (Figure 4.1.7) gives a more detailed visual representation of the model to classify landslide pixels. As the matrix indicates, the model was able to identify 12,290,348 pixels of No Landslide (true negatives) and 315,303 pixels of Landslide (true positives), which means that it did well at identifying both the background and the landslide areas. This shows that the model can be used to reveal important spatial patterns and be able to discriminate between stable terrain and landslide-influenced regions. However, the confusion matrix also highlights certain misclassifications. There were 148,032 non-landslide pixels that were false positives (incorrectly classified as landslide), and 287,981 landslide pixels that were false negatives (incorrectly classified as non-landslide). The comparatively large value of false negatives suggests that the model does not identify some areas of landslides, which could be as a result of fine changes in textures or complicated structures. Moreover, the asymmetry between the background and landslide pixels influences the sensitivity of the model and makes the model more biased towards the majority class.

📊 Classification Report:

	precision	recall	f1-score	support
No Landslide	0.98	0.99	0.98	12438380
Landslide	0.68	0.52	0.59	603284
accuracy			0.97	13041664
macro avg	0.83	0.76	0.79	13041664
weighted avg	0.96	0.97	0.96	13041664

Figure 4.1.8: Performance Metrics of SegFormer

Figure. 4.1.8 provides the pixel-wise evaluation metrics, which gives a quantitative measure of the model performance. The model has a cumulative accuracy of 96.66, which means that most of the pixels are correctly identified. A 0.6805 value in the landslide detection indicates that the model is right about 68% of the times when it predicts a landslide. This indicates a fairly good capability of not showing false positives when locating areas of landslides. Nevertheless, the recall value of 0.5226 shows that the model only identifies approximately 52% of the real landslide pixels, which implies that much of the landslides areas are overlooked. The F1-score of 0.5912 shows a trade-off between precision and recall, which means that the overall performance of the landslide class is moderate. Moreover, the Intersection over Union (IoU) score of 0.4197 and Dice coefficient of 0.5912 indicates a medium overlapping between the anticipated and actual landslide areas. These findings demonstrate that although the model

works well in general, it should be improved further to increase the recall and segmentation accuracy in detecting landslides.

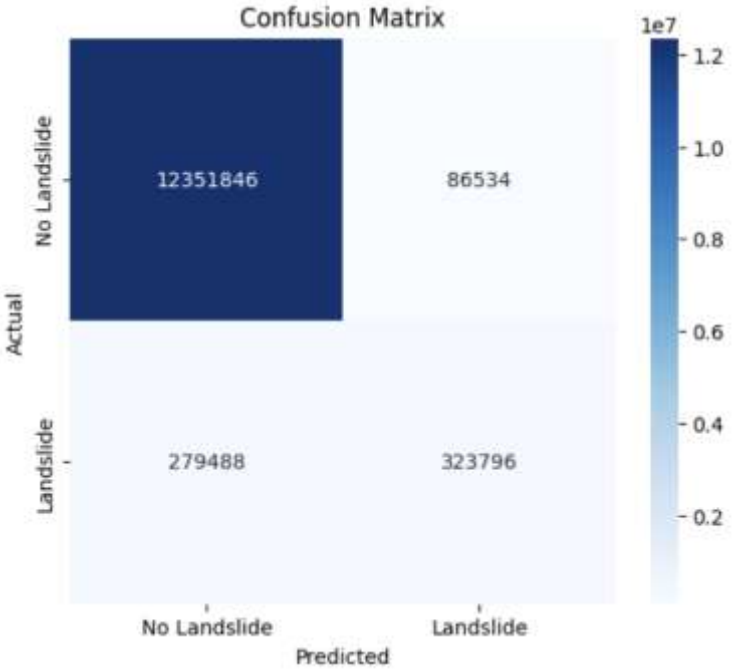

Figure 4.1.9: Confusion Matrix of MobileViT

The confusion matrix (4.1.9) shows how the model performs at the pixel level in terms of classifying between landslide and non-landslide areas. This model accurately identifies No Landslide (true negatives) and Landslide (true positives) with 12,351,846 and 323,796 pixels respectively, which is a good model in distinguishing between background and affected areas. This indicates that the model is able to capture spatial features and patterns that are related to landslides and is also highly accurate in the most frequent background class. The high count of accurately localized background pixels helps in boosting the overall performance of the model. However, the confusion matrix also reveals some misclassifications that impact performance. The non-landslide pixels that are falsely predicted as landslide are 86,534 (false positives), and the landslide pixels that are falsely detected as non-landslide are 279,488 (false negatives). The fact that there were more false negatives means that the model fails to detect some landslides, especially small, piece-meal, or less pronounced ones. It means that the model is quite effective in general, but its sensitivity to the identification of all pixels of landslides can be enhanced. To improve the effectiveness of the model in real-world applications like hazard detection and risk assessment, the errors need to be addressed.

```
📊 Classification Report:
              precision    recall  f1-score   support

No Landslide       0.98      0.99      0.99  12438380
   Landslide       0.79      0.54      0.64    603284

    accuracy                           0.97  13041664
   macro avg       0.88      0.76      0.81  13041664
weighted avg       0.97      0.97      0.97  13041664
```

Figure 4.1.10: Performance Metrics of MobileViT

Figure 4.1.10 shows the pixel-based assessment values, which give a detailed evaluation of the segmentation performance of the model. The model has a total accuracy of 97.19, which means that most pixels are rightly categorized. This value of 0.7891 indicates that the model predicts a landslide accurately almost 79 times out of 100, indicating a very high capability of minimizing the number of false positives. This means that it is more reliable in determining the areas in which landslides occur as compared to previous models. The recall value of 0.5367 however means that only approximately half of the real landslide pixels are captured and thus, a portion of landslides are missed. The F1-score of 0.6389 gives the landslide class a moderate level of performance in terms of precision and recall. Moreover, the Intersection over Union (IoU) value of 0.4694 and Dice coefficient of 0.6389 indicate that there is a reasonable overlap between the predicted and actual landslide areas. These findings indicate that even though the model has high overall accuracy and enhanced precision, additional improvement is necessary to enhance recall and attain more accurate and complete landslide detection.

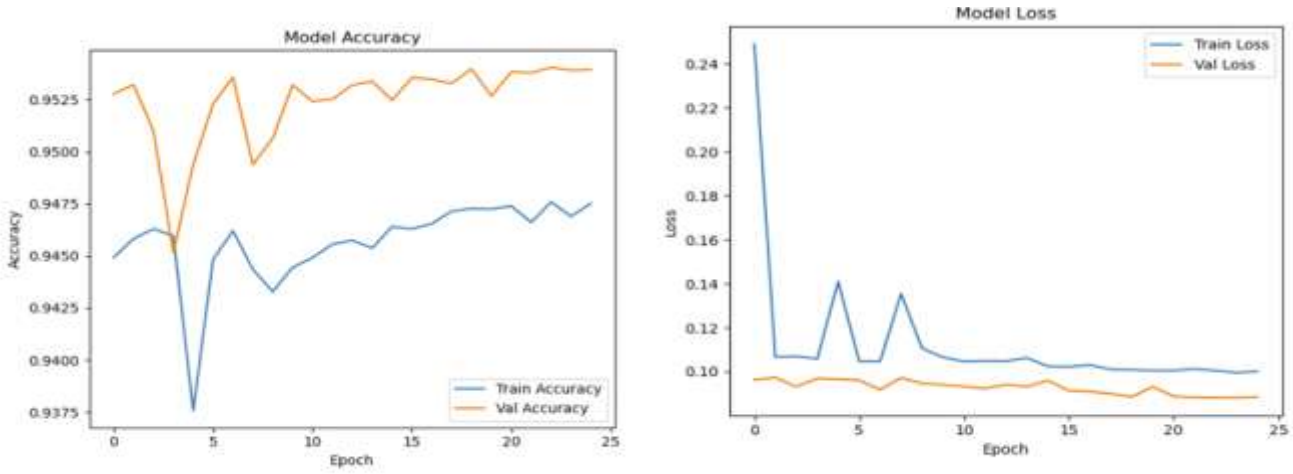

Figure-4.1.11: Model Loss and Accuracy Graph for U-Net

Model Accuracy:

The validation accuracy curve and training curve of the U-Net model show a gradual and consistent increase with each epoch, which shows successful learning during the training process. At the beginning, the model experiences small oscillations as it starts to discover the underlying patterns in the data, and then the oscillations start to decrease gradually in subsequent epochs. The validation accuracy is always higher than the training accuracy, with the accuracy being in the range of about 95.3, indicating that the model can be generalized to unseen data. The above behavior implies that the model is not overfitting and can capture significant spatial and contextual characteristics of the input images. The convergent continuity in the curves is also an indication of stable convergence and effective learning of segmentation patterns.

Model Loss:

The loss curves of the training and validation shows that there is an evident downward trend which is evidence that the model is essentially minimizing the prediction errors with time. The loss in training regularizes significantly in the initial epochs, which means that the model can acquire such properties as edges and landslide area textures in a relatively short time frame. The loss values will be inclined towards a constant in the course of training and this is a pointer of the refinement of the model parameters. The loss on the validation is also always less than the training loss, which shows that it has a good generalization property and minimal overfitting. Such consistency means that the U-Net model is dependable and can provide accurate segmentation results even in different epochs.

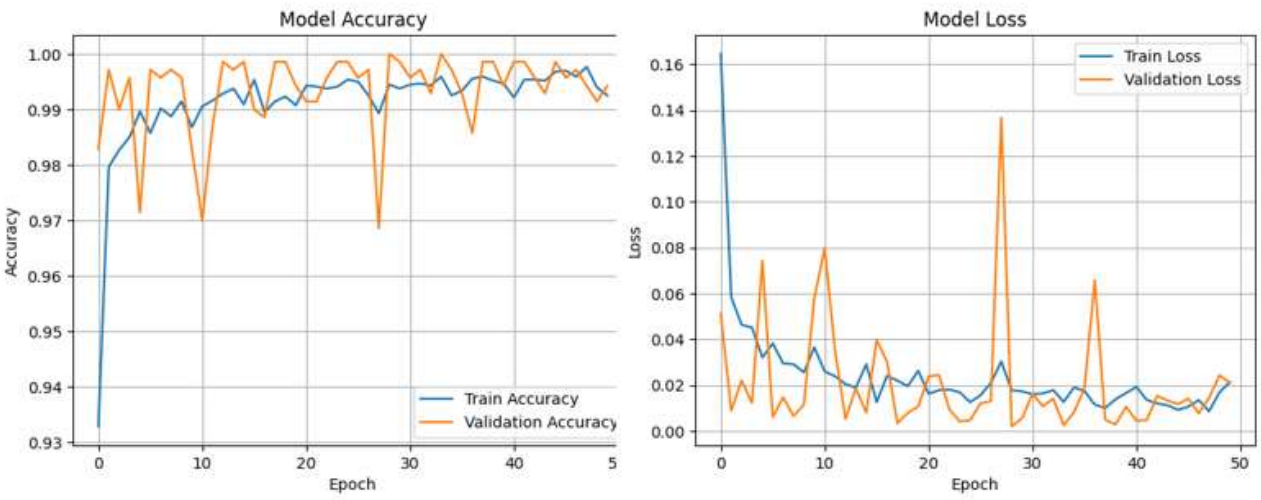

Figure-4.1.12: Model Loss and Accuracy Graph for DeepDeconvResU-Net

Model Accuracy:

The accuracy curve shows the learning performance of the Attention U-Net model as the model gains accuracy with the training epochs. The training and validation accuracy exhibit a steady positive trend, which implies successful feature learning and consistent generalization. The validation accuracy starts with a higher value than training accuracy during the initial epochs, which is frequent when the model is quick to adapt to broader patterns on the validation set. Training accuracy increases steadily at the beginning of training, and eventually the training and validation accuracies stabilize both at about 96.25 and 96.30 percent, respectively, at the last epoch. The high correlation between both curves is an indication that both curves are not overfitting or underfitting and have high generalization potential with unseen data.

Model Loss:

The loss curve also illustrates convergence and stability of training of the model. The loss in the training dwindles very fast at the first few epochs, indicating speedy optimization and efficient weight changes. The gradual decrease occurs over the next few epochs, but it eventually reaches a stability of about 0.06, which means that the model has acquired the relevant spatial features that are used to perform segmentation correctly. The validation loss follows the same trend of decreasing and is always lower than the training loss thereby indicating that the model is generalizing and not memorizing the training data. Small changes in the validation loss are also anticipated in segmentation tasks because of the difference between the shapes of landslides and the complexity of the background, although, the general downward trend indicates that the model has converged successfully and reached a steady performance.

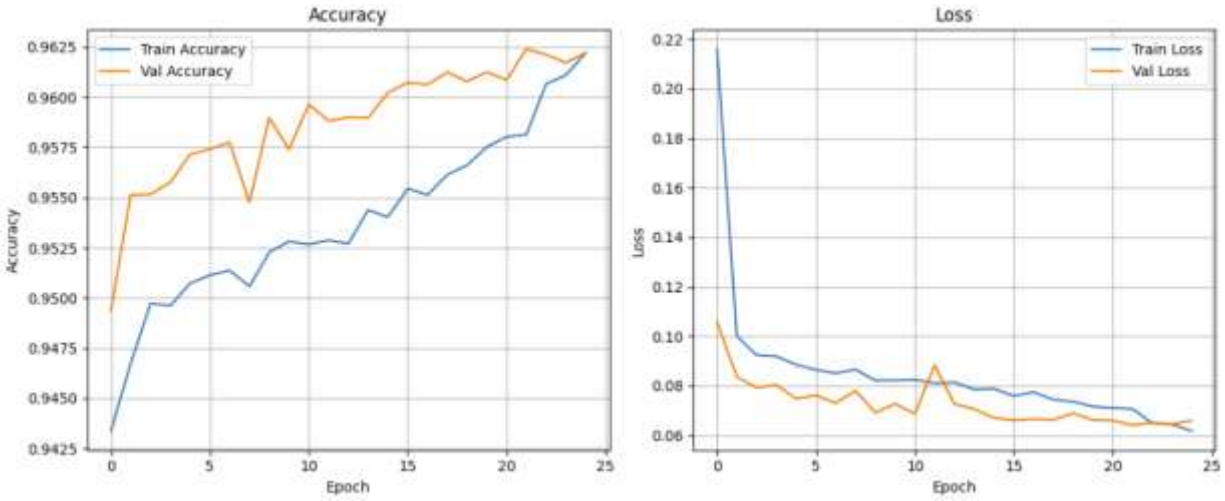

Figure 4.1.13: Model Accuracy and Loss Graph f AttentionU-Net

Model Accuracy:

The validation accuracy curve and training curve of the SegFormer model show that the learning trend is fast and stable with the number of epochs. First, the accuracy of the training is rapidly rising with a lower value, which means that the model rapidly learns significant spatial and contextual details about the input satellite images. Training and validation accuracy also increases steadily and levels off as training continues, to around 95 per cent or 96 per cent, a sign of steady learning behaviour. The validation accuracy is consistently a bit larger than or extremely near the training accuracy during the process, indicating that the model can generalize well to the unknown data without any substantial overfitting. This can be attributed to the transformer-based encoder of SegFormer, which encodes the global dependencies and long-range relationships within the image. Compared to the classical CNN-based models, SegFormer builds the attention mechanism to learn local and global patterns, thus being very useful in the complex segmentation tasks such as landslide detection. The fact that the curves of accuracy are converging smoothly shows that the model is stable and can derive meaningful representations of the dataset. In general, the SegFormer model demonstrates good generalization and consistent performance in pixel-wise segmentation.

Model Loss:

The training and validation loss curves of the SegFormer model exhibit a significant decrease during the initial epochs, indicating efficient learning and rapid convergence. The loss decreases drastically initially when the model acquires more basic characteristics like variations of texture and demarcation of regions with landslides. As the training progresses, both curves gradually reduce and level off, indicating that the model is adjusting the

parameters to minimize the prediction errors. The gradual decrease in loss demonstrates a well-organized training process without any instability. The loss in validation is always a little bit less than the training loss, which is a good sign of good generalization and low overfitting. This behaviour demonstrates the efficiency of the lightweight decoder and transformer encoder pair in SegFormer that enables effective feature extraction and reconstruction. The fact that the loss curve does not have large fluctuations or spikes also supports the idea that training dynamics are stable. In general, the trend of the losses shows that SegFormer is a sound and effective model, which can obtain precise and consistent segmentation outcomes in landslide detection tasks.

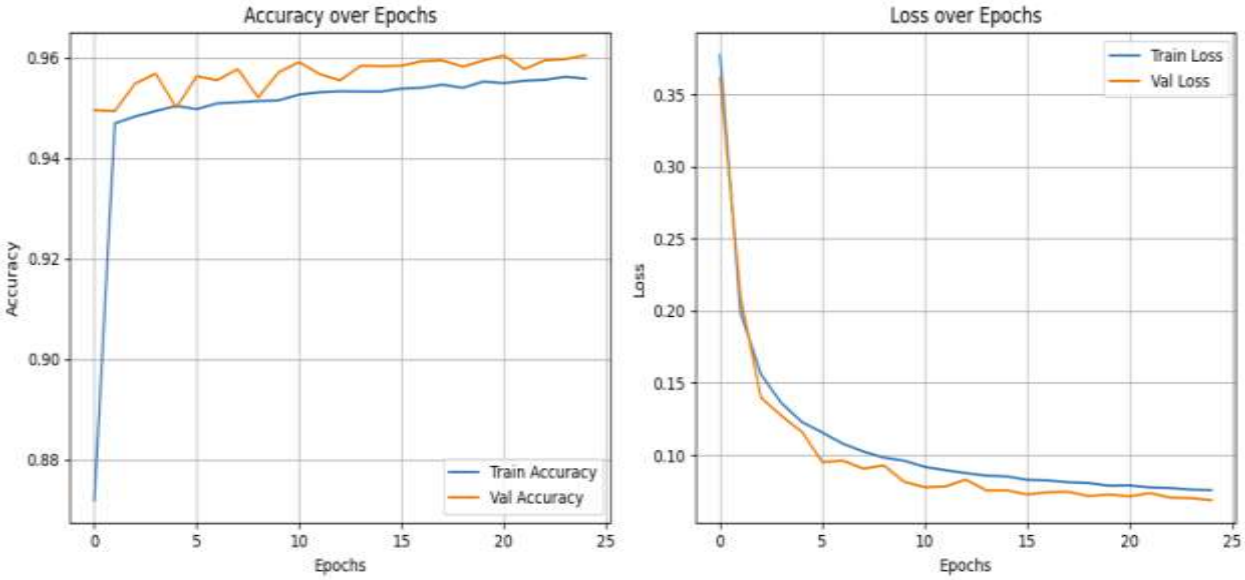

Figure 4.1.14: Model Accuracy and Loss Graph of SegFormer

Model Accuracy:

The training and validation accuracy curve of the MobileViT model indicates that the model is learning effectively as the curves are steadily increasing with the epochs. The training accuracy is initially at a relatively lower level but it rapidly rises as the model continues to extract the necessary spatial features in the input images. The training and validation accuracy curves increase with training and level off at 96%96.5, indicating a steady convergence. The validation accuracy is a little bit better than the training accuracy in most of the epochs, this indicates that the model is generalized to the unseen data without major overfitting. This can be explained by the fact that MobileViT has a hybrid architecture that uses convolutional layers to extract local features and transformer blocks that are lightweight to extract global context. These two components are combined to help the model learn the fine-grained details and long-range dependencies in the image in an efficient manner. The accurate curves of the

smooth and stable curves are evidence to the fact that the model is strong and can deal with the changes in patterns of landslides and terrain conditions. In general, MobileViT can provide a decent compromise between performance and computational efficiency, which makes it a good choice in the context of segmentation in the real world.

Model Loss:

The training and validation loss curves of the model MobileViT exhibit a steep reduction in the first few epochs, which means that the model learned quickly and able to extract features. This rapid decrease in loss demonstrates that the model is able to detect significant features like edges, textures and region boundaries that are related to landslides very fast. The loss values approach a steady value as training proceeds, indicating that the model is adjusting its parameters to reduce prediction errors and enhance the quality of the segmentation. The curves are smooth, which means that there is no serious change in the training. The validation loss is always small compared to the training loss, which is a good sign that shows good generalization and low overfitting. It is anticipated that the validation curve may have occasional minor variations owing to changes in validation samples but this does not have any significant impact on the overall trend. MobileViT combines convolutional operations with attention based on transformers to enable the efficient learning of features at lower computational complexity. On the whole, the trends of losses prove that the MobileViT model is stable, efficient, and can provide credible performance in the landslide segmentation tasks.

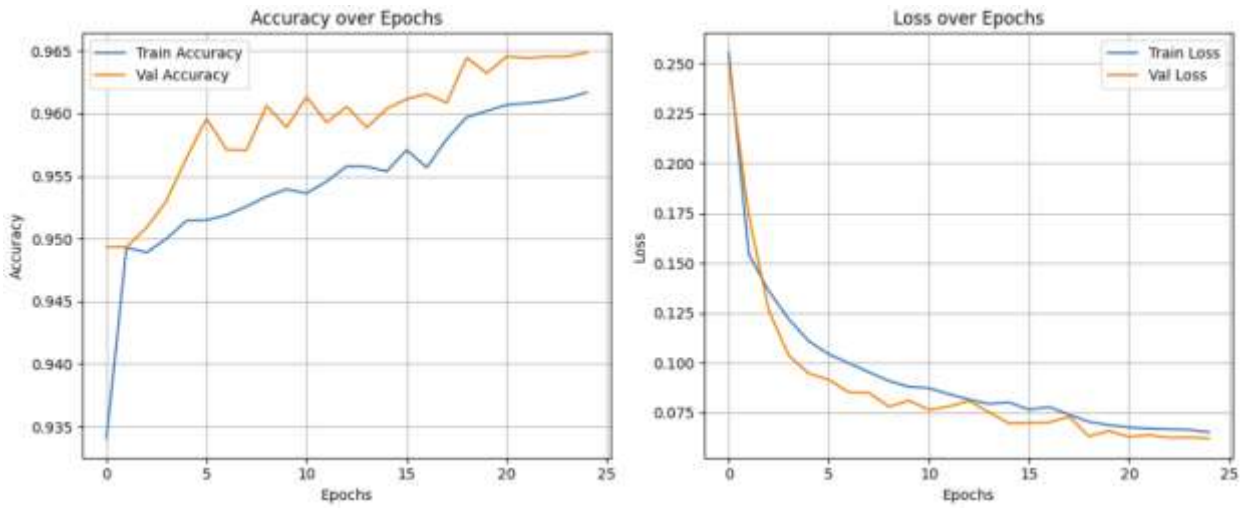

Figure 4.1.15: Model Accuracy and Loss Graph of MobileViT

Model Accuracy:

The curves of training and validation accuracy of the MobileViT model depict that the model

is successfully learning, as the accuracy is steadily increasing with the epochs. The training accuracy is relatively low at the first stage but rapidly grows when the model starts to extract critical features of space in the input images. Training and validation accuracy curves increase gradually with training and stabilize at 96%-96.5 corresponding to constant convergence. The validation accuracy is somewhat higher than the training accuracy during most of the epochs, indicating that the model can be generalized effectively to unknown data, with minimal overfitting. This achievement is due to the hybrid nature of MobileViT that consists of convolutional layers to extract local features and small transformer blocks to capture global information. By combining these two elements, the model is able to learn fine-grained details and long-range dependencies in the image effectively. The curves of smooth and stable accuracy are strong indicators showing that the model is resilient and can accommodate changes in landslide patterns and terrain conditions. In general, MobileViT has a decent tradeoff between performance and computation efficiency, which makes it applicable to the real-world segmentation.

Model Loss:

As seen in the training and validation loss curves, the MobileViT model has a sharp decrease in the first epochs, which means that it learns quickly and extracts the features successfully. This rapid reduction in loss demonstrates the capacity of the model to recognize, in a short time, significant features like edges, textures, and boundary of regions that are related to landslides. With further training, the loss values decrease and become flat, indicating that the model is adjusting its parameters to reduce prediction errors and enhance the quality of segmentation. The fact that the curves are smooth shows that there is no extreme change in training. The validation loss is always kept very slightly less than the training loss, and this is a good sign of good generalization and little overfitting. The validation curve may have slight changes occasionally because of the variation in the validation samples, but they do not cause major deviations in the general trend. Convolutional operations and transformer-based attention enable the MobileViT to learn features efficiently with low computational complexity. Altogether, the loss trends validate that the MobileViT model is stable, efficient, and can provide reliable performance in the landslide segmentation tasks.

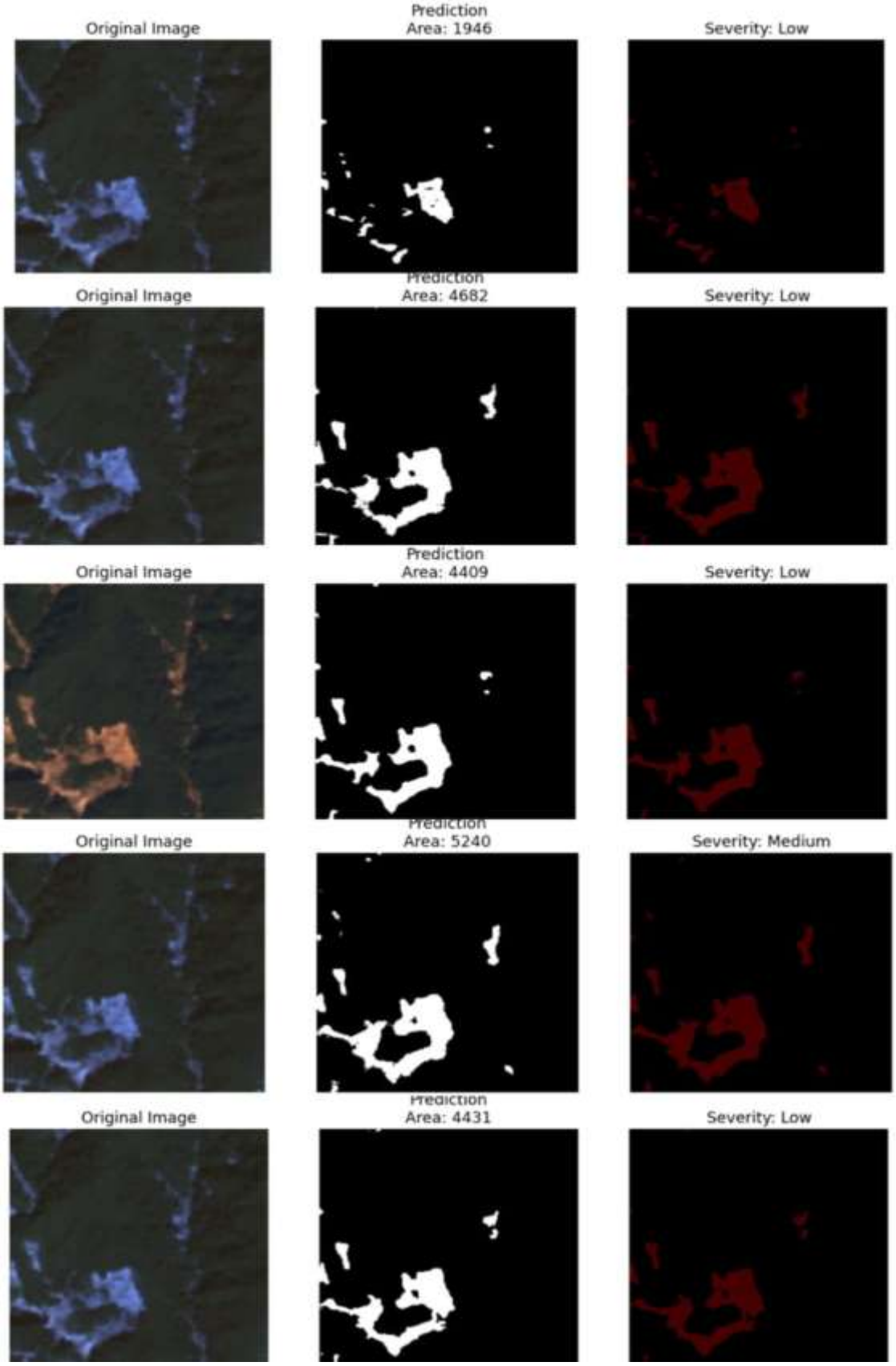

Figure 4.1.16: Prediction, Area Calculation, and Severity Analysis of all Models

(a) UNet, (b) DeepDeconvResUNet, (c) Attention UNet, (d) SegFormer and (e) MobileViT

The qualitative results shown in Figure 4.1.16 represent the segmentation outputs of five different models: U-Net, DDR U-Net, Attention U-Net, SegFormer, and MobileViT. Every row represents a particular model and shows the original satellite image, the predicted segmentation mask and the severity-based visualization of the predicted landslide area.

The U-Net model (Figure a) can be used to identify major landslides but the boundaries of the segmentation are relatively coarse. It captures the overall landslide form but some smaller areas are either missed or seem to be broken up. The relative smaller predicted areas mean that the severity classification of this model mainly lies under the Low category.

DDR U-Net (Figure b) shows better segmentation than U-Net with a better continuity and reconstruction of landslide areas. The residual connections are included, which assists in capturing more detailed features, leading to more complete segmentation masks and the slightly higher values of the area estimation.

Attention U-Net model (Figure c) adds another layer of segmentation of quality by concentrating on the relevant regions via attention gates. It minimizes the background noise and enhances boundary accuracy particularly in rough terrain. The estimated masks are more sophisticated and consistent with real landslide areas than the earlier models.

SegFormer (Figure d) demonstrates good performance in capturing global contextual information. It generates a more coherent and organized segmentation results, especially with large landslide zones. This is evident in increased area estimation (e.g., 5240 pixels), resulting in Medium severity classification. SegFormer is more proficient with large and intricate patterns.

A balanced model MobileViT (Figure e) offers a combination of lightweight convolution and transformer features. It is efficient in segmentation, with fairly good boundary detection and is computationally efficient. It is a bit less detailed as compared to SegFormer, but it yields credible predictions and severity classification.

In general, each of the models is effective at identifying landslide areas, although their accuracy varies in the boundary accuracy, completeness and area estimation. Transformer-based models such as SegFormer and MobileViT demonstrate better contextual knowledge, whereas hybrid CNN-based models such as DDR U-Net and Attention U-Net are effective at spatial localization. The combination of area calculation and severity classification also makes the system more useful in practical applications in real-world landslide risk assessment.

4.2 RESULT ANALYSIS

Model Summary:

Several pixel-wise segmentation metrics, such as Accuracy, Precision, Recall, F1-Score, Intersection over Union (IoU), and Dice Coefficient, are used to evaluate the performance of different deep learning models, i.e., U-Net, DDR U-Net, Attention U-Net, SegFormer, and MobileViT. These measures in table 4.2.1 give a holistic view of the potential of each model in determining landslide and non-landslide areas. DDR U-Net has the highest overall performance with the highest accuracy of 97.30, and higher F1-Score (0.6740), IoU (0.5083) and Dice Coefficient (0.6740). This implies that DDR U-Net is able to strike the right balance of accuracy and recall hence it is very appropriate when it comes to identifying the location of landslides. Residual connections and deep deconvolution layers contribute to propagating features and retaining the spatial information, resulting in the high quality of segmentation.

Another model that demonstrates competitive results is the MobileViT model with an accuracy of 97.19% and the highest precision (0.7891) of all models. This implies that MobileViT can be very useful in reducing the false positives and recognizing the landslide pixels correctly. Its hybrid design, consisting of convolutional operations coupled with lightweight transformer blocks, allows it to efficiently learn local and global features. Its recall (0.5367) is however a notch lower compared to DDR U-Net, which means that there are still areas of landslides that are not covered. Nevertheless, MobileViT is a good trade-off between performance and computation efficiency and is thus suitable to real-time and resource-constrained applications.

Attention U-Net model is an enhancement of the simple U-Net that adds attention modules which assist in focusing the image on the significant areas. It has an accuracy of 97.15 and higher recall (0.5757) than U-Net meaning that it is able to detect landslide pixels better. Its general results regarding the F1-score (0.5861) and the IoU (0.4145) remain less than DDR U-Net and MobileViT, though. The attention mechanism selects features better, but may be unable to obtain local complex global dependencies that are there in large scale satellite images. Likewise, the SegFormer model, a transformer architecture-based model, demonstrates good performance with an accuracy of 96.66 and mediocre scores on other metrics. It is highly efficient in capturing global contextual data through self-attention mechanisms, although its simplified form of operation and limitations of computation can limit the potential of it in this project.

Conversely, the simple U-Net model achieves the lowest score on all models, especially in recall (0.3757), IoU (0.2945), and Dice Coefficient (0.4549). This means that although U-Net

does a good job of detecting background areas, it is not good at detecting smaller and complex regions of landslides. This hinders its capability to record complex patterns and boundaries due to the absence of advanced feature enhancement mechanisms. This underscores the need to adopt better architectures like residual networks, attention models, and transformer-based models to perform more accurate segmentation tasks.

On balance, the comparative analysis obviously indicates that advanced and hybrid architectures are better than traditional models at landslide segmentation. DDR U-Net is the most successful model, with a good balance of accuracy, precision, recall, and overlap-based measurements.

MobileViT model also demonstrates competitive performance with an accuracy of 97.19% and the highest precision (0.7891) of all the models. This indicates that MobileViT can be used successfully to reduce the number of false positives and detect landslide pixels correctly. The local and global feature learning is effectively achieved by its architecture (hybrid) that consists of convolutional operations and lightweight transformer blocks.

MobileViT and SegFormer show the efficiency of transformer-based methods, especially in capturing the global context, and Attention U-Net can be improved by focused learning of features. These findings indicate that the use of a combination of convolutional and transformer-based methods could dramatically improve the results of segmentation. Future work can involve correcting the class imbalance, larger datasets, and more efficient transformer architectures to make better detection accuracy and robustness in real world use.

Table 4.2.1 Result Analysis for models

Metric	U-Net	DDR U-Net	Attention U-Net	SegFormer	MobileViT
Accuracy	0.9684	0.9730	0.9715	0.96666	0.9719
Precision	0.5766	0.7805	0.5970	0.6805	0.7891
Recall	0.3757	0.5931	0.5757	0.5226	0.5367
F1-Score	0.4549	0.6740	0.5861	0.5912	0.6389
IoU	0.2945	0.5083	0.4145	0.4197	0.4694
Dice Coefficient	0.4549	0.6740	0.5861	0.5912	0.6389

4.3 OBSERVATIONS FROM THE WORK

Performance of Basic U-Net Model

The U-Net model demonstrates a decent overall accuracy, but is not effective at detecting landslide areas. It works well in marking non-landslide (background) pixels, but has low recall and IoU scores, meaning that large numbers of landslide pixels are missed. This is attributed to the fact that it is not very effective in capturing the complex and irregular spatial patterns in the satellite images.

Improvement with Advanced CNN Models

DDR U-Net and Attention U-Net models are highly effective in enhancing the performance of segmentation. The overall performance of DDR U-Net is the highest because there are residual connections that improve feature learning and gradient flow. Attention U-Net enhances the concentration on pertinent areas, which result to the detection of landslides areas of the basic U-Net.

Effectiveness of Transformer-Based Models

Transformer-based models such as SegFormer and hybrid models such as MobileViT demonstrate good ability to capture global contextual information. SegFormer is effective at interpreting big scale terrain patterns whereas MobileViT is highly accurate at incorporating CNN and transformer features. These models show the significance of learning global features in the segmentation tasks.

Effect of Class Imbalance

The proportion of non-landslide pixels in the dataset is significantly large in comparison to the landslide pixels. Such an uneven distribution of classes causes high accuracy and low recall on landslide detection. Most of the models are more likely to predict the background and omit certain areas of landslides which also underscores the importance of balancing imbalanced data.

Importance of Hybrid Architectures

Hybrid models, which are based on CNN and transformer methods, are superior to individual models. DDR U-Net and MobileViT demonstrate that the accuracy and robustness of segmentation can be improved by using both local feature extraction and contextual understanding, particularly in the case of complex terrain images.

Stability of Model Training

Most models exhibit a steady learning pattern with little or no overfitting as demonstrated by the training and validation curves. A validation accuracy is often near or greater than training accuracy, signifying good generalization. There is smooth convergence on loss curves which indicates effective training

Landslide Detection problems.

In most cases, landslide regions are small, irregular and look like the adjacent areas, a fact that

makes it hard to identify them. This results in false negative and decreased recall. Identifying such minute differences is one of the greatest problems in all models.

Effectiveness of Area and Severity Analysis

The model is also more practical due to the combination of area calculation and severity classification. The system offers valuable insights to disaster management and risk assessment by estimating the area affected and classifying the severity (Low, Medium).

Overall Model Comparison

DDR U-Net is the best of all models in terms of balanced metrics F1-score, IoU, and Dice coefficient. MobileViT and SegFormer have promising results, whereas U-Net is the least effective. This is an indication of the need to adopt advanced architectures to enhance the performance of segmentation.

CHAPTER-5

CONCLUSION AND FUTURE STUDY

5.1 CONCLUSION

This study presents a comprehensive analysis of deep learning and transformer-based architectures for landslide segmentation using remote sensing imagery from the Landslide4Sense dataset. The models evaluated in this work include U-Net, Deep Deconvolutional Residual U-Net (DDR U-Net), Attention U-Net, SegFormer, and MobileViT. All models demonstrated the capability to perform pixel-wise classification of landslide and non-landslide regions, highlighting the effectiveness of deep learning techniques in automated geospatial hazard detection. The integration of preprocessing, segmentation, area estimation, and severity classification further enhances the practical applicability of the proposed system.

Among the evaluated models, DDR U-Net achieved the best overall performance, with superior results in terms of F1-score, IoU, and Dice coefficient, indicating its strong ability to balance precision and recall while preserving fine spatial details. The use of residual connections and deep deconvolution layers significantly improved feature extraction and boundary reconstruction. MobileViT also showed competitive performance, achieving high precision and accuracy by effectively combining convolutional and transformer-based feature learning. This demonstrates the advantage of hybrid architectures in capturing both local and global features efficiently.

The SegFormer model, based on transformer architecture, successfully captured global contextual information and long-range dependencies within the input images. Although its performance was slightly lower than DDR U-Net and MobileViT in this implementation, it demonstrated strong potential for handling large-scale and complex terrain variations. Attention U-Net improved segmentation performance compared to the baseline U-Net by focusing on relevant regions, while the basic U-Net model showed comparatively lower performance, particularly in detecting smaller landslide regions.

In addition to segmentation, the proposed system incorporates area calculation and severity classification, which provide meaningful insights into the extent and impact of landslides. This transforms the model from a simple segmentation tool into a practical decision-support system for disaster management. Overall, the results confirm that advanced and hybrid deep learning models significantly improve landslide detection accuracy and robustness. The study highlights the potential of combining CNN-based and transformer-based approaches for developing efficient and reliable landslide monitoring systems.

5.2 Future Study

Full Implementation of Transformer Architectures:

The current implementation of SegFormer and MobileViT can be further enhanced by using their complete architectures with pre-trained weights. This can significantly improve performance by leveraging large-scale learned representations and deeper transformer layers.

Integration of Multi-Modal Data:

Future research can incorporate additional data sources such as Digital Elevation Models (DEM), Synthetic Aperture Radar (SAR), and rainfall data. Combining multi-modal inputs can improve model performance, especially in challenging terrains and weather conditions.

Development of Lightweight and Real-Time Models:

Optimizing models like MobileViT for real-time deployment can make the system more practical for early warning systems. Reducing computational complexity and memory usage will allow deployment on edge devices and low-resource environments.

Integration with Early Warning Systems:

The proposed model can be integrated with real-time monitoring systems and GIS platforms to provide automated alerts and support disaster management authorities in making timely decisions.

Expand Testing on Diverse Regions:

Test the models on different geographical locations to improve generalization and robustness

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APPENDIX

```python
def unet_model(input_size=(IMAGE_SIZE, IMAGE_SIZE, 3)):
    inputs = Input(input_size)
    # Encoder
    c1 = Conv2D(64, 3, activation='relu', padding='same')(inputs)
    c1 = Conv2D(64, 3, activation='relu', padding='same')(c1)
    p1 = MaxPooling2D()(c1)
    c2 = Conv2D(128, 3, activation='relu', padding='same')(p1)
    c2 = Conv2D(128, 3, activation='relu', padding='same')(c2)
    p2 = MaxPooling2D()(c2)
    c3 = Conv2D(256, 3, activation='relu', padding='same')(p2)
    c3 = Conv2D(256, 3, activation='relu', padding='same')(c3)
    p3 = MaxPooling2D()(c3)
    c4 = Conv2D(512, 3, activation='relu', padding='same')(p3)
    c4 = Conv2D(512, 3, activation='relu', padding='same')(c4)
    p4 = MaxPooling2D()(c4)
    # Bottleneck
    c5 = Conv2D(1024, 3, activation='relu', padding='same')(p4)
    c5 = Conv2D(1024, 3, activation='relu', padding='same')(c5)
    # Decoder
    u6 = Conv2DTranspose(512, 2, strides=2, padding='same')(c5)
    u6 = concatenate([u6, c4])
    c6 = Conv2D(512, 3, activation='relu', padding='same')(u6)
    c6 = Conv2D(512, 3, activation='relu', padding='same')(c6)
    u7 = Conv2DTranspose(256, 2, strides=2, padding='same')(c6)
    u7 = concatenate([u7, c3])
    c7 = Conv2D(256, 3, activation='relu', padding='same')(u7)
    c7 = Conv2D(256, 3, activation='relu', padding='same')(c7)
    u8 = Conv2DTranspose(128, 2, strides=2, padding='same')(c7)
    u8 = concatenate([u8, c2])
    c8 = Conv2D(128, 3, activation='relu', padding='same')(u8)
    c8 = Conv2D(128, 3, activation='relu', padding='same')(c8)
    u9 = Conv2DTranspose(64, 2, strides=2, padding='same')(c8)
    u9 = concatenate([u9, c1])
    c9 = Conv2D(64, 3, activation='relu', padding='same')(u9)
    c9 = Conv2D(64, 3, activation='relu', padding='same')(c9)
    outputs = Conv2D(1, 1, activation='sigmoid')(c9)
```

Figure A1: Code Snippet of U-Net Model

```python
def conv_block(x, filters):
    x_skip = x
    x = Conv2D(filters, 3, padding='same')(x)
    x = BatchNormalization()(x)
    x = Activation('relu')(x)
    x = Conv2D(filters, 3, padding='same')(x)
    x = BatchNormalization()(x)
    if x_skip.shape[-1] != filters:
        x_skip = Conv2D(filters, 1, padding='same')(x_skip)
    x = Add()([x, x_skip])
    x = Activation('relu')(x)
    return x
def encoder_block(x, filters):
    x = conv_block(x, filters)
    p = MaxPooling2D((2, 2))(x)
    return x, p
def decoder_block(x, skip, filters):
    x = Conv2DTranspose(filters, (2, 2), strides=(2, 2), padding='same')(x)
    x = concatenate([x, skip])
    x = conv_block(x, filters)
    return x
def build_ddrunet(input_shape=(IMAGE_SIZE, IMAGE_SIZE, 3)):
    inputs = Input(shape=input_shape)
    s1, p1 = encoder_block(inputs, 64)
    s2, p2 = encoder_block(p1, 128)
    s3, p3 = encoder_block(p2, 256)
    s4, p4 = encoder_block(p3, 512)
    b1 = conv_block(p4, 1024)
    d1 = decoder_block(b1, s4, 512)
    d2 = decoder_block(d1, s3, 256)
    d3 = decoder_block(d2, s2, 128)
    d4 = decoder_block(d3, s1, 64)
    outputs = Conv2D(1, (1, 1), padding='same', activation='sigmoid')(d4)
    return Model(inputs, outputs)
```

Figure A2: Code Snippet of DeepDeconvResU-Net Model

```python
# Step 3: Attention Gate
def attention_gate(x, g, inter_channels):
    theta_x = Conv2D(inter_channels, 1, padding='same')(x)   # skip connection
    phi_g = Conv2D(inter_channels, 1, padding='same')(g)
    #  ◆ Resize gating signal
    if theta_x.shape[1] is not None and phi_g.shape[1] is not None:
        if theta_x.shape[1] != phi_g.shape[1]:
            scale_h = theta_x.shape[1] // phi_g.shape[1]
            scale_w = theta_x.shape[2] // phi_g.shape[2]
            phi_g = UpSampling2D(size=(scale_h, scale_w), interpolation="bilinear")(phi_g)
    add_xg = Add()([theta_x, phi_g])
    act = Activation('relu')(add_xg)
    psi = Conv2D(1, 1, padding='same')(act)
    psi = Activation('sigmoid')(psi)
    return Multiply()([x, psi])
```

```python
def attention_unet(input_size=(IMAGE_SIZE, IMAGE_SIZE, 3)):
    inputs = Input(input_size)
    # Encoder
    c1 = Conv2D(64, 3, activation='relu', padding='same')(inputs)
    c1 = Conv2D(64, 3, activation='relu', padding='same')(c1)
    p1 = MaxPooling2D()(c1)
    c2 = Conv2D(128, 3, activation='relu', padding='same')(p1)
    c2 = Conv2D(128, 3, activation='relu', padding='same')(c2)
    p2 = MaxPooling2D()(c2)
    c3 = Conv2D(256, 3, activation='relu', padding='same')(p2)
    c3 = Conv2D(256, 3, activation='relu', padding='same')(c3)
    p3 = MaxPooling2D()(c3)
    c4 = Conv2D(512, 3, activation='relu', padding='same')(p3)
    c4 = Conv2D(512, 3, activation='relu', padding='same')(c4)
    p4 = MaxPooling2D()(c4)
    # Bottleneck
    c5 = Conv2D(1024, 3, activation='relu', padding='same')(p4)
    c5 = Conv2D(1024, 3, activation='relu', padding='same')(c5)
    # Decoder with Attention
    g4 = Conv2D(512, 1, activation='relu', padding='same')(c5)
    att4 = attention_gate(c4, g4, 256)
    u6 = Conv2DTranspose(512, 2, strides=2, padding='same')(c5)
    u6 = concatenate([u6, att4])
    c6 = Conv2D(512, 3, activation='relu', padding='same')(u6)
    c6 = Conv2D(512, 3, activation='relu', padding='same')(c6)
    g3 = Conv2D(256, 1, activation='relu', padding='same')(c6)
    att3 = attention_gate(c3, g3, 128)
    u7 = Conv2DTranspose(256, 2, strides=2, padding='same')(c6)
    u7 = concatenate([u7, att3])
    c7 = Conv2D(256, 3, activation='relu', padding='same')(u7)
    c7 = Conv2D(256, 3, activation='relu', padding='same')(c7)
    g2 = Conv2D(128, 1, activation='relu', padding='same')(c7)
    att2 = attention_gate(c2, g2, 64)
    u8 = Conv2DTranspose(128, 2, strides=2, padding='same')(c7)
    u8 = concatenate([u8, att2])
    c8 = Conv2D(128, 3, activation='relu', padding='same')(u8)
    c8 = Conv2D(128, 3, activation='relu', padding='same')(c8)
    g1 = Conv2D(64, 1, activation='relu', padding='same')(c8)
    att1 = attention_gate(c1, g1, 32)
    u9 = Conv2DTranspose(64, 2, strides=2, padding='same')(c8)
    u9 = concatenate([u9, att1])
    c9 = Conv2D(64, 3, activation='relu', padding='same')(u9)
    c9 = Conv2D(64, 3, activation='relu', padding='same')(c9)
    outputs = Conv2D(1, 1, activation='sigmoid')(c9)
    return Model(inputs, outputs, name="Attention_U-Net")
```

Figure A3: Code Snippet of AttentionU-Net Model

```python
# 🧱 Step 5: SegFormer Model
def transformer_block(x, num_heads=4, key_dim=64):
    attn = tf.keras.layers.MultiHeadAttention(num_heads=num_heads, key_dim=key_dim)(x, x)
    x = Add()([x, attn])
    x = LayerNormalization()(x)

    ffn = Dense(key_dim * 2, activation="relu")(x)
    ffn = Dense(key_dim)(ffn)
    x = Add()([x, ffn])
    x = LayerNormalization()(x)
    return x
def build_segformer(input_shape=(IMAGE_SIZE, IMAGE_SIZE, 3)):
    inputs = Input(shape=input_shape)
    # Patch embedding
    x = Conv2D(64, 7, strides=4, padding='same')(inputs)
    x = BatchNormalization()(x)
    x = Activation('relu')(x)
    h, w, c = x.shape[1], x.shape[2], x.shape[3]
    # Flatten
    x = Reshape((h * w, c))(x)
    # Transformer encoder
    for _ in range(4):
        x = transformer_block(x)
    # Reshape back
    x = Reshape((h, w, c))(x)
    # Decoder
    # Decoder (FIXED)
    x = Conv2DTranspose(128, 3, strides=2, padding='same')(x)  # 64 → 128
    x = BatchNormalization()(x)
    x = Activation('relu')(x)
    x = Conv2DTranspose(64, 3, strides=2, padding='same')(x)   # 128 → 256
    x = BatchNormalization()(x)
    x = Activation('relu')(x)
    outputs = Conv2D(1, 1, activation='sigmoid')(x)
    return Model(inputs, outputs, name="SegFormer")
```

Figure A4: Code Snippet of SegFormer Model

```python
def build_mobilevit_unet(input_shape=(IMAGE_SIZE, IMAGE_SIZE, 3)):
    inputs = Input(shape=input_shape)
    # Encoder (NO MobileViT here)
    c1 = conv_block(inputs, 64)
    p1 = MaxPooling2D()(c1)
    c2 = conv_block(p1, 128)
    p2 = MaxPooling2D()(c2)
    c3 = conv_block(p2, 256)
    p3 = MaxPooling2D()(c3)
    c4 = conv_block(p3, 256)
    p4 = MaxPooling2D()(c4)
    # Bottleneck
    b = conv_block(p4, 256)
    b = mobilevit_block(b, 256)
    # Decoder
    u1 = Conv2DTranspose(256, 2, strides=2, padding="same")(b)
    u1 = concatenate([u1, c4])
    c5 = conv_block(u1, 256)
    u2 = Conv2DTranspose(256, 2, strides=2, padding="same")(c5)
    u2 = concatenate([u2, c3])
    c6 = conv_block(u2, 256)
    u3 = Conv2DTranspose(128, 2, strides=2, padding="same")(c6)
    u3 = concatenate([u3, c2])
    c7 = conv_block(u3, 128)
    u4 = Conv2DTranspose(64, 2, strides=2, padding="same")(c7)
    u4 = concatenate([u4, c1])
    c8 = conv_block(u4, 64)
    # Output
    outputs = Conv2D(1, 1, activation="sigmoid")(c8)
    return Model(inputs, outputs, name="MobileViT_UNet")
```

Figure A5: Code Snippet of MobileViT Model

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